STAINED GLASS WINDOWS

SECTION UIP 6

GENERAL INFORMATION

Stained glass windows are in the fine arts field. Many windows, classified by age or beauty as having antique or historical value, or designed by named artists, must be valued as art objects by fine arts specialists.

The majority of stained glass which is installed today is the work of skilled artisans working in shops or studios who normally put a square foot price on their craft. The following average costs will enable the assessor to estimate the replacement cost or proposed cost within reasonable limits. Glass materials, design and studio prominence vary considerably and costs may vary from a plus 70% to minus 20%.

These costs are based on the least dimension of the dominant size of pieces making up the total window. A window, most of whose pieces of glass are 3" wide, would be priced from the 2" to 4" column, whether 3", 6" or 12" long.

Price all windows by the area of the enclosing rectangle.

INSTALLATION

Installation costs run from 21.15 to 52.50 per square foot, depending on geographical location, height from ground, and size and shape of window. For stained glass overlay, costs run 38.25 to 70.50 per square foot including materials.

RESIDENTIAL TYPE, STRAIGHT SEAMS

(Cost per Square Foot)

TYPE	DOMINANT SIZE OF INDIVIDUAL PIECES			
	14" and up	12"	8"	under 6"
Clear glass	51.50	59.50	68.00	79.00
Single color	73.50	84.00	96.00	108.00
Multi–color	81.00	90.00	102.00	114.00

Rondels and accidental designs cost 102.00 to 221.00 per sq. ft. Blown clear restoration window glass cost 20.05 to 37.50 per sq. ft. For single glazed over, add 8.56 to 11.80; for double glazed; 20.35 to 42.75 per sq. ft.

SIMULATED ART GLASS

Acrylic resin panels cost 104.00 to 190.00 per square foot. For leading on two sides add 28.75 per square foot.

Art glass	4" and up	3"	under 2"
Plain mosaics, regular pieces, few embellishments	102.00	124.00	152.00
Contemporary abstract	119.00	183.00	281.00
Minimum painted, fired field and borders (Figures 1 and 2)	147.00	206.00	267.00
Simple designs, geometric patterns silhouettes, etc. (Figure 3)	190.00	334.00	595.00
Simple figures and scenes, little detail	380.00	565.00	830.00
Highly detailed figures and scenes (Figure 4)	830.00	1270.00	1920.00

ILLUSTRATIONS



FIGURE 1

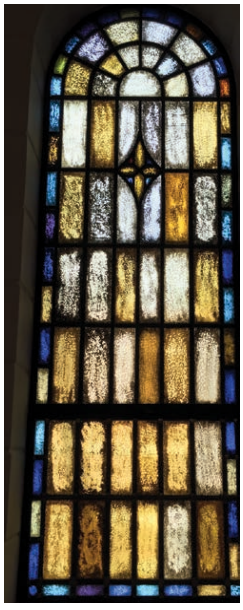


FIGURE 2



FIGURE 3



FIGURE 4

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STAINED GLASS WINDOWS

SECTION UIP 6

SIMULATED ART GLASS

(Cost Per Square Foot)

TYPE	DOMINANT SIZE OF INDIVIDUAL PIECES		
Art glass (continued)	**4" and up**	**3"**	**under 2"**
Rose or wheel windows, intricate designs	665.00	1100.00	1870.00
Detailed figures, large decorative background areas (Figure 5)	710.00	1150.00	1920.00
Faceted glass, 1" or less thickness, cast in epoxy, contemporary abstract	206.00	289.00	411.00
Faceted glass, 1" or less thickness, cast in epoxy, figures and scenes (Figure 6)	259.00	365.00	488.00
Faceted glass, 1-1/2", bound in concrete, add 20% to epoxy cost.			
Ceiling domes, stock design up to 12' diameter including soffit, cost 1080.00 to 1830.00 per square foot of horizontal projection.			
Etched and Beveled glass			
Minimum sand-etched design,			
1/4" plate	65.50	72.00	76.00
tempered	72.50	77.50	84.00
Simple figures and scenes,			
1/4" plate	133.00	149.00	168.00
tempered	143.00	161.00	183.00
Intricate designs,			
1/4" plate	236.00	305.00	396.00
tempered	252.00	312.00	404.00

For stock designpiece assemblies, deduct 20%.
For obscure glass, add 10%. Multilayered glass, add 15% to 35%.
For triple glazing between clear safety glass, add 73.50 to 88.50 per square foot.

ILLUSTRATIONS

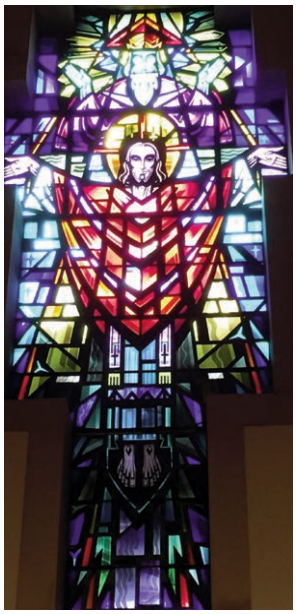


FIGURE 5



FIGURE 6

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STOREFRONTS
SECTION UIP 6

GENERAL INFORMATION

The costs in this section are for use in computing storefronts with the Unit-in-Place and/or Segregated Cost Methods. The Calculator Costs include fronts commensurate with the quality and type of building.

The floor and ceiling areas in the display and vestibule are normally included in the building shell and are not part of the storefront, however, the display platforms and dropped display and vestibule ceilings as well as special floors are part of the front. The basic wall above the front is normally part of the building shell except for interior malls, but the ornamentation is properly computed with the storefront. Some times the display front is a very small part of the entire front elevation of the building and the major portion is better computed from the regular wall costs in Section UIP 5 or SEG 3 adding any front ornamentation.

If the cost of replacing an old or damaged storefront is desired, the costs of removing the old front and making any necessary structural changes must be added to the cost of the new front.

The following text and pictures illustrate some examples of estimating storefronts, while the detailed example on Page 5 covers the complete estimate.

GOOD DEPARTMENT STORE



The storefront to be computed includes the glassed areas, entrances, marquees and bulkheads. The wall to the sides of the storefront is not included since it is identical to, and priced with, the balance of the walls. The added ornamentation on this area should be priced as part of the storefront.

When computing the area of the basic storefront, include the entrances and sides of the window areas but not the bulkhead.

In the illustration above, the basic storefront is good quality: aluminum anodized tube window frame and tempered plate glass. The wall framing above windows/doors includes: steel framed, insulation, vapor barrier, gypsum board/plywood, Marble cladding.

Additional ornamental finish consists of Marble cladding around jamb (sides) and header (top) of entrance area wall opening. The lighting consists of both average and good spotlights and up-lighting, all lights in the window and marquee should be counted.

The entrance doors are clear anodized aluminum tube frame, tempered glass, push/pull hardware. The canopy is good-quality aluminum faced on a cantilevered steel frame.

Typical cost per linear foot of storefront elevation =$1,970 to $2,480 (Including display front to top of marquee only.)

EXCELLENT RETAIL STORE



The largest proportion of the front of this building is an ordinary E.I.F.S. (Exterior Insulation Finish System) wall, with a smaller pr portion for window and door openings, and should be priced with the balance of the walls of the building. The window and doors are an average bronze aluminum tube frame and tempered plate glass for display and entrance areas.

Ornamentation finish consists of native stone over the front pillars, corners and bottom 3’ of the exterior walls.

The entrance doors are good quality 1/2” tempered plate glass balanced doors and hardware.

The vestibule ceiling is a dropped ceiling with plaster finish and the vestibule floor is good marble which may be priced from Section UIP 2.

There is no special lighting for the front except in the entrance

Typical cost per front foot of building facade = $2,775 to $3,300 (Including entire front wall of building.)

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STOREFRONTS
SECTION UIP 6

GOOD RETAIL STORE



In this storefront, most of the wall is built of E.I.F.S. (Exterior Insulation Finishing System) which may be priced with the balance of the building walls, leaving only the windows, awnings and entrance to be priced from the storefront costs.

The windows and doors are bronze-framed plate glass with average spotlights for lighting. Display platforms are carpeted. The display and vestibule ceilings are average drywall, and the sides of the entrance may be priced from the bulkhead walls and ornamentation.

Other ornamentation consists of granite faced pillars and the extended curved parapet wall at the top of the building.

Typical cost per front foot of building facade = $1,220 to $1,610
(Including entire front wall of building.)

AVERAGE MALL STORE



This basic mall storefront is predominately glass and has some minimal wood trim. The sign area is a basic metal, painted, single face with an aluminum frame. The average plain galvanized steel roll-up grille is priced separately and is not included as an additive to the basic storefront area. The shallow display platforms with drywall ceiling and spotlights are all of average quality.

Typical cost per front foot of store entrance = $540 to $855
(Including entire front wall of mall.)

AVERAGE RETAIL STORE



This basic storefront is average plate glass and aluminum framed entrance and windows. The front columns are imitation and the front gable sign area is E.I.F.S. (Exterior Insulation Finishing System)

There is a small metal and steel frame marquee with minimal lighting.

Typical cost per front foot of building facade = $665 to $ 905
(Including entire front wall of building.)

LOW-COST RETAIL STORE



This basic storefront has plain brick with minimal arches, ¼" plate glass windows and an average aluminum frame, canvas covered canopy.

The ornamentation is low quality local brick and workmanship.

There is no front lighting and the entrance doors are low cost aluminum with ¼" plate glass. There are no other items to price

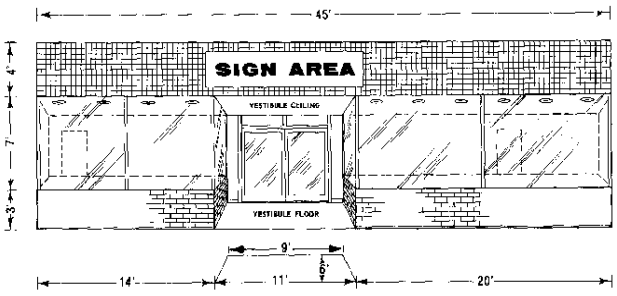
Typical cost per front foot of building facade = $426 to $595
(Including entire front wall of building.)

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STOREFRONTS
SECTION UIP 6

EXAMPLE

The storefront illustrated is for a typical, small, average retail store and has display platforms and dropped ceilings in the display area and vestibule. The vestibule floor and the basic wall above the front were included in the cost of the building shell. Costs are found on the following pages.



COMPUTATION

Basic storefront area: (46 x 7) + (9 x 10)			
= 412 sq. ft. aluminum trim, average	@	50.50	= 20,806
Basic bulkhead area: (14 + 6 + 6 + 20) x 3			
= 138 sq. ft. brick masonry, average	@	30.75	= 4,244
Ornamentation area: 45 x 4 (sign area)			
= 180 sq. ft. ceramic tile, average	@	23.50	= 4,230
Lighting: 10 recessed spots, average	@	289.00	= 2,890
Entrance doors:			
2 aluminum, 1/4" glass, average	@	2480.00	= 4,960
Display platform area: 6 x (14+15) + 6 x (20+21)			
2 2			
210 sq. ft., 36"-high display, average	@	13.65	= 2,867
210 sq. ft. carpet, average, medium grade (Section UIP 2)	@	6.27	= 1,317
Display ceiling area, including vestibule, 45 x 6 = 270 sq. ft. dropped plaster ceiling, average	@	10.10	= 2,727
Display back area: 7 x (15 + 21)			
252 sq. ft. single wall plywood, avg	@	9.37	= 2,361
Total storefront cost			46,402

Cost per linear foot of building facade = $1,031

BASIC STOREFRONT

Cost per square foot of actual front including door and glass area, but not bulkhead or sign areas. For display islands and bays use -area of all sides. Costs include glass.

	LOW COST	AVG.	GOOD
Wood trim	25.75	33.00	42.25
Aluminum trim	37.50	50.50	67.50
Stainless steel or bronze trim	62.00	81.00	108.00
Structural glazing, no trim	100.00	141.00	206.00
Greenhouse or curved glass, aluminum	74.50	90.00	111.00

Double glazed add 40%.
Interior nonweatherized mall front, deduct 10% to 20%.

BASIC BULKHEAD WALLS

	LOW COST	AVG.	GOOD
Wood frame	8.40	10.55	13.25
Steel frame	8.56	12.05	16.95
Concrete block masonry, standard	13.30	17.15	22.40
add for face block	1.87	3.09	6.03
Concrete, formed	20.75	27.00	34.50
Concrete, precast or tilt up	15.15	18.85	23.65
add for textured finishes	1.72	3.67	7.19
Brick masonry	24.95	30.75	38.00
add for face brick	3.82	4.82	6.03

ORNAMENTATION

Costs per square foot of ornamented area.

Brick veneer	15.15	19.90	25.75
Simulated veneer or panel, 1/4" - 3/8"	13.25	17.70	23.35
Ceramic tile	19.15	23.50	29.00
*Mosaic tile	28.25	38.00	50.50
Glass block, white or aqua	55.00	64.00	73.50
Metal panels or screens	24.45	36.50	54.50
Preformed wall panels	4.48	10.50	21.30
Local stone, ashlar or panel	43.75	56.50	72.00
Rubble or rustic	33.25	43.50	55.00
Imitation veneer	14.30	21.60	31.50
Marble or granite	57.00	77.50	108.00
Terra cotta	39.50	50.50	65.00
Stucco	6.28	7.10	8.10
On masonry	4.55	5.54	6.60
add for synthetic			
On rigid insulation	3.33	4.82	7.10
*built–up (EIFS) panel relief	7.10	9.59	15.70
Wood, plywood	3.67	5.13	7.26
Shingles or shakes	4.89	6.28	7.91
Shingles, decorative panels	5.88	7.00	8.56
Siding	4.55	6.60	8.80
Ornamented molding or trim	14.65	23.65	37.50
add for treated wood	0.74	0.88	1.31
*Note - For pictorial artwork, add	15.10	25.25	42.25
Carved brick, add	214.00	259.00	320.00

LIGHTING

Add for each storefront fixture in display areas, marquees, etc.

Std. open incandescent fixtures	113.00	144.00	190.00
Recessed or adj. incandescent	168.00	289.00	488.00
Fluorescent, open	312.00	471.00	685.00
Fluorescent, diffused	404.00	585.00	810.00
Illuminated exit, incandescent	161.00	281.00	488.00

Emergency light packs, see Section UIP 3.

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STOREFRONTS
SECTION UIP 6

ENTRANCE DOORS

Cost per door, including glazing, to be added to cost of basic front.

	LOW COST	AVG.	GOOD
Wood and glass	1100.00	1500.00	1940.00
Aluminum frame, 1/4'' glass	1870.00	2480.00	3250.00
Bronze or stainless steel, 1/4'' glass	4225.00	5250.00	6500.00
Tempered glass, 1/2''	3525.00	4150.00	4725.00
3/4''	3875.00	5050.00	6500.00
Sliding glass, automatic two–way entry	21500.00	25800.00	31700.00
Revolving door & cage (7' dia. base)	45500.00	63250.00	87750.00
Bronze or stainless steel	70500.00	92750.00	123000.00
add or deduct 15% for each foot variation.			
Automatic door opener, add.	7800.00	9300.00	11500.00
Balanced door, add	7200.00	10500.00	15000.00
Panic hardware, add	875.00	1100.00	1440.00
Mall fronts, per sq. ft. of opening, excluding basic front cost.			
Chain or scissor closures, horizontal.	24.00	33.00	45.75
Grilles, roll up, plain aluminum or steel	45.25	58.50	77.50
Bronze anodized or mirror glazed	57.00	73.00	96.00
Ornate, stainless steel	70.50	90.00	117.00
For horizontal operation, deduct 10% add for motorized operation, each	1690.00	2340.00	3325.00
Sliding glass panels, aluminum	36.25	49.75	66.50
Bronze, anodized	45.25	61.00	82.50

DISPLAY PLATFORMS

Cost per square foot of display floor area. Add for finish flooring from Section UIP 2.

12'' high.	8.40	11.35	15.45
24'' high.	9.05	12.30	16.65
36'' high.	10.10	13.65	18.70

DISPLAY CEILINGS

Cost per square foot of dropped ceilings not already included in main store ceiling, exclusive of lighting.

Acoustic tile, organic fiber, wood or cane	6.03	7.48	9.19
Mineral fiber, fiberglass	5.95	8.40	11.50
Gypsum board, taped and painted	6.20	7.48	9.05
Mirror–faced panels	29.00	35.50	43.50
Plaster, standard	8.10	10.10	12.85
Acoustical or Keene's	8.65	11.00	14.15
Plastic panels, translucent	11.40	14.90	19.15
Wood, decorative	12.05	22.55	42.25
Plywood, softwood or Hardboard	7.00	8.80	11.40
Hardwood	9.12	15.45	25.75

DISPLAY BACKS

Cost per square foot of back, including doors.

Masonry	13.50	19.00	27.00
Metal channels, 2-1/4'' gypsum board	15.55	17.15	19.15
Single wall, plywood	6.92	9.37	12.90
Steel studs, 2'' solid plaster	16.35	18.40	20.75
Wood studs, finished both sides	9.37	13.00	17.95

AWNINGS

Metal, enameled, per sq. ft.	20.05	25.25	31.75
Fabric, per front foot	214.00	281.00	351.00
Prefab economy units	97.00	134.00	183.00

MARQUEES

Cost per square foot of marquee.

	LOW COST	AVG.	GOOD
Metal, ornamented, steel frame	54.00	69.50	88.50
Metal, plain, steel frame	39.00	54.00	73.00
Metal, plain, wood frame	36.25	49.75	68.00
Wood or stucco, wood frame	32.25	43.75	60.50

For light false-mansard structures, deduct 50%. For greater detail, see Section UIP 4, Page 5.

SIGNS

Signs normally are not a part of the building. They are separately purchased and often are the property of the tenant. Most signs are custom made and costs are highly variable. The costs per square foot below represent national medians and are for signs hung or mounted flush on buildings without extra structure or foundation. See Section UIP 14 for pole costs and service station signs. For oval signs use area of enclosed rectangle. Interior non-weatherized signs may cost 5% to 15% less.

SIGN TYPE	AREA IN SQUARE FEET < 20	20 – 50	50 – 100	> 100
Illuminated plastic,				
Single face	252.00	197.00	161.00	130.00
Double faced	358.00	289.00	221.00	175.00
Without illumination, deduct.	37.50	28.00	22.10	17.95
Metal, painted, single face	94.50	84.00	74.50	65.50
Double faced	119.00	107.00	94.50	82.50
Porcelainized, add per face	20.05	18.25	15.10	14.30
Wood, painted, single face.	44.25	39.50	35.50	32.25
Double faced	57.00	51.00	45.75	41.50
Carved, plain, single face	259.00	228.00	206.00	175.00
Double faced	396.00	334.00	281.00	228.00
Ornate artwork, add up to	40%	45%	50%	55%
Sandblasted or Routed, deduct.	12%	14%	17%	20%
Neon tubing, add per face				
plain	45%	40%	40%	35%
Ornate lettering	70%	65%	60%	55%
Projected wall bracket				
supports, add	17.95	13.90	11.00	8.56
Rotators, add	84.00	74.50	65.00	57.00

Individual letters, cost per inch of height	COST RANGE		
Masonite or foamboard	5.22	–	7.91
Plastic	6.60	–	9.99
Wood	11.00	–	14.30
Steel	11.90	–	17.15
Aluminum	17.95	–	28.75
Stainless steel	23.20	–	33.75
Bronze	25.75	–	36.50
Illuminated, Plexiglas, metal frame	27.75	–	42.00
Stainless steel or porcelainized frame	43.50	–	54.50
Without illumination, deduct	6.60	–	8.65
Window lettering, permanent.	1.64	–	3.02
Gold leaf	19.15	–	46.75
Carved in stone (raised letters, add 100%)	5.22	–	7.26
Time and temperature, 18'' 24'' display, cost per set			
Alternating, single face	17500.00	–	23400.00
Double faced	25900.00	–	35700.00
Continuous display, add.	7800.00	–	12000.00
Electronic message, 18"24" single display, per lin. ft.	4825.00	–	6050.00
Double faced	5300.00	–	7200.00

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STONEWORK
SECTION UIP 6

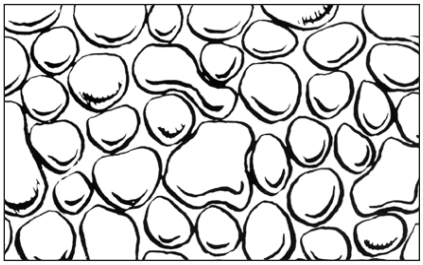
GENERAL INFORMATION

There are a number of quarries throughout the United States and Canada producing various types of building stone. The distance, typically 100 miles, from the quarries affects the cost of any of these, but most areas have one or two types of stone which are commonly used for facing or walls because of their local availability. Costs will vary by coloring, graining, texture and finish. Large quantities of matched stone or intricate jobs requiring much cutting and fitting can cost twice the averages as shown.

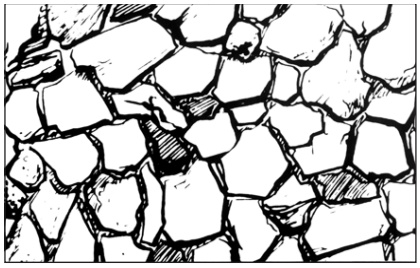
The following costs are typical ranges per square foot for locally available competitive stone, in place. For stone veneer on wood or steel studs, add 2.57 to 5.82 per square foot. Dry bed (hand placed) local fieldstone walls will cost 50% to 100% more. For stone carving, add 125.00 to 202.00 per surface square foot; hand work add 100%. Works designated as having historical or artistic merit must be valued by fine arts specialists.

RUSTIC STONE

COBBLE

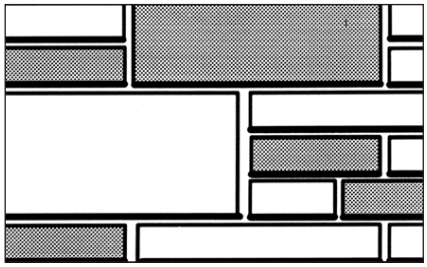


RUBBLE – FIELDSTONE

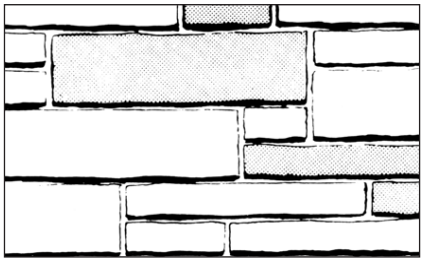


ASHLAR FACING

COURSED SAW BED



RANDOM ROUGH BED



LOCAL STONE

Cost per sq. ft., Sandstone, Brownstone, Quartzite, etc.

	LOW COST	AVG.	GOOD
Ashlar veneer, 4” – 6”, rough	32.75	42.25	55.00
Sawed bed	38.25	49.75	65.00
Facing panels, 1’’	39.25	53.50	73.50
Native cobblestone veneer, 4’’ – 6’’	25.50	33.00	43.00
Rubblestone	29.50	38.25	49.00
Cobblestone walls, 12’’	28.00	33.00	38.25
Each additional 6’’	8.01	8.55	9.08
Rubblestone walls, 12’’	36.50	44.00	51.00
Each additional 6’’	10.20	10.70	11.25
Rough–cut block, per cubic foot	62.00	80.50	104.00
Miscellaneous trim, per cubic foot	85.00	117.00	157.00

LIMESTONE

Ashlar veneer, 4” – 6”, rough	39.25	48.25	60.00
Sawed bed	46.50	60.00	79.50
Facing panels, 1’’	44.75	60.00	83.50
Rubble veneer, 4’’ – 6’’	32.25	39.25	49.75
Rough–cut block, per cubic foot	71.50	89.50	115.00
Miscellaneous trim, per cubic foot	100.00	141.00	195.00

GRANITE

Ashlar veneer, 4” 6”, rough	44.00	55.50	70.50
Polished	53.50	69.00	88.00
Facing panels, 1”	57.50	73.00	91.00
2”	64.00	80.50	102.00
4”	75.00	93.00	115.00
Rubble veneer, 4” 6”	38.25	46.50	57.50
Rough–cut block, per cubic foot	65.00	91.00	125.00
Trim, polished, per cubic foot.	172.00	239.00	343.00
Trim or steps, sawn, per cubic foot	98.50	114.00	131.00

MARBLE

Marble costs can be divided into four groups, according to the degree of color variation and structure:

GROUP A

Sound marbles with a smooth uniform surface requiring no waxing or filling, best suited for exterior walls, with little variation in texture and color. These are almost entirely local marbles and are strongly competitive in price.

GROUP B

Similar to Group A but with less favorable working qualities and a little more variation in color and texture. A few marbles in this group are imported.

GROUP C

Uncertain variations in working qualities, large flaws, voids, striations, and good color variation. Most imported marbles will fall in this category or Group D.

GROUP D

Marbles similar to Group C, but including those stones with maximum variations in working qualities, texture, and color. This group contains many of the highly colored and variegated marbles which are most prized for ornamentation.

FACING PANELS

(Cost per square foot)

GROUP	LOW COST	AVG.	GOOD
A	47.25	60.00	79.50
B	57.50	73.00	91.00
C	69.00	85.00	107.00
D	83.50	102.00	125.00

SLATE

Facing panels, 1"	43.00	57.00	80.50

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ARCHITECTURAL COLUMNS

SECTION UIP 6

GRANITE COLUMNS

Cost per linear foot of height measured from bottom of base to top of capital. The costs of architectural granite columns will vary a great deal with the distance from quarry and mill, and the following typical cost ranges should be increased where the stone must be crated for shipment over long distances or for foreign granite. For carved stone cladding, reduce costs by 50%.

DIAMETER OF SHAFT AT BASE	ROUND COST RANGE	FLUTED COST RANGE
8''	232.00 – 298.00	471.00 – 505.00
12''	329.00 – 404.00	605.00 – 675.00
16''	471.00 – 545.00	800.00 – 900.00
20''	635.00 – 755.00	1070.00 – 1180.00
24''	865.00 – 1040.00	1410.00 – 1600.00
28''	1180.00 – 1410.00	1900.00 – 2060.00
32''	1640.00 – 1960.00	2500.00 – 2750.00
36''	2350.00 – 2600.00	3425.00 – 3650.00
40''	3225.00 – 3650.00	4550.00 – 4875.00
44''	4475.00 – 4925.00	6050.00 – 6500.00

BASES AND CAPITALS

Cost per base or capital to be added to total column cost. The capital forms the top most member of a column.

COLUMN DIAMETER	BASE AND CAPITALS PLAIN	BASE AND CAPITALS MODERATE	CAPITALS ORNATE
8''	820.00	1600.00	2875.00
12''	1330.00	2480.00	4650.00
16''	1900.00	3375.00	6250.00
20''	2480.00	4175.00	7950.00
24''	3075.00	5100.00	9750.00
28''	3750.00	6000.00	11500.00
32''	4325.00	7000.00	13400.00
36''	5000.00	7850.00	15100.00
40''	5700.00	8850.00	17000.00
44''	6400.00	9750.00	19000.00

ORNAMENTAL CONCRETE COLUMNS

Costs are given per linear foot of architectural nonbearing column measured from the bottom of the base to the top of the capital. If a base and capital are used the additional cost of each may be added to the total column cost. The diameter of the column is measured at the bottom of the column above the base.

DIA.	TYPE OF COLUMN			BASE		CAPITAL	
OF BASE	ROUND	FLUTED	SQUARE	PLAIN	ORNATE	PLAIN	ORNATE
12''	61.50	100.00	75.00	128.00	187.00	284.00	655.00
14''	79.50	120.00	94.50	150.00	225.00	359.00	770.00
16''	96.00	141.00	109.00	179.00	261.00	404.00	920.00
20''	130.00	179.00	148.00	261.00	389.00	570.00	1300.00
24''	172.00	225.00	195.00	389.00	545.00	835.00	1840.00
28''	209.00	277.00	239.00	545.00	765.00	1180.00	2525.00
32''	254.00	321.00	284.00	770.00	1130.00	1640.00	3600.00
36''	307.00	373.00	343.00	1130.00	1600.00	2360.00	5000.00

WOOD STAVE COLUMNS

Costs are given per linear foot of standard architectural columns measured from the bottom of the base to the top of the capital. The diameter of the column is measured at the bottom of the column above the base.

DIA. OF SHAFT AT BASE	ROUND COST RANGE		SQUARE COST RANGE	
	PLAIN	FLUTED	PLAIN	FLUTED
6''	57.50 – 63.50	66.00– 74.00	60.50 – 84.00	64.50 – 90.50
8''	79.50 – 95.00	98.00– 115.00	93.50 – 129.00	95.00 – 131.00
12''	135.00 – 165.00	165.00– 203.00	158.00 – 227.00	165.00 – 234.00
16''	196.00 – 248.00	241.00– 300.00	234.00 – 338.00	241.00 – 360.00
20''	256.00 – 345.00	322.00– 419.00	322.00 – 457.00	330.00 – 495.00
24''	322.00 – 443.00	405.00– 540.00	419.00 – 595.00	428.00 – 645.00
28''	367.00 – 525.00	488.00– 640.00	505.00 – 755.00	525.00 – 810.00
32''	436.00 – 625.00	570.00– 775.00	615.00 – 900.00	645.00 – 980.00
36''	505.00 – 755.00	670.00– 915.00	730.00 – 1050.00	755.00 – 1160.00

BASES AND CAPITALS

Cost per each base or capital to be added to total column cost.

COLUMN DIAMETER	BASE AND CAPITALS PLAIN	BASE AND CAPITALS MODERATE	CAPITALS ORNATE
6''	95.00	292.00	436.00
8''	110.00	345.00	540.00
12''	165.00	495.00	755.00
16''	234.00	700.00	1040.00
20''	338.00	1020.00	1490.00
24''	488.00	1410.00	2090.00
28''	700.00	2030.00	2875.00
32''	1020.00	2875.00	4050.00
36''	1480.00	4050.00	5600.00

For solid composite fiberglass and marble chip columns and bases deduct 30% to 50% from the above wood costs. For architectural interior plaster columns, add 100%.

ORNAMENTAL ALUMINUM COLUMNS

Costs are given per linear foot of architectural nonbearing column measured from the bottom of the base to the top of the capital. Add for each base or capital. The diameter of the column is measured at the bottom of the column above the base.

DIAMETER OF SHAFT AT BASE	FLUTED ROUND	FLUTED SQUARE	BASE AND CAPITALS PLAIN	BASE AND CAPITALS ORNATE
4''	17.85	19.90	41.75	181.00
6''	19.90	26.75	57.50	285.00
8''	34.75	45.75	72.50	398.00
10''	52.50	71.50	90.50	505.00
12''	74.50	101.00	101.00	615.00
14''	99.50	137.00	115.00	755.00
16''	126.00	181.00	129.00	860.00
20''	196.00	-------	158.00	1100.00
24''	270.00	-------	181.00	1320.00
30''	405.00	-------	218.00	1720.00

For hard coated, cored polystyrene columns, deduct 20% to 30% from the aluminum costs above.

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COLD STORAGE
SECTION UIP 7

INSTRUCTIONS

To estimate the total cost of a cold-storage plant, determine the cost of the basic building; then add the cost of insulation, equipment, and doors as indicated on this page. Since these components represent a large proportion of the total cost, it is advisable to determine accurately the temperatures maintained, and thickness and type of insulation. The table to the right is a guide to the determination of insulation thickness for maintaining various temperatures.

TYPICAL INSULATION REQUIREMENTS

These typical thicknesses are often varied to suit the requirements of a specific installation. Some of the items which may affect the amount of insulation and refrigeration used are: the types of commodity being stored and the frequency of removal and replacement of goods being refrigerated.

The table is based upon corkboard, styrofoam, foamglas, and fiberglass board. For mineral wool batts, increase thickness by 40%. For urethane and isocyanurate insulation, decrease thickness by 40%.

ROOM TYPE	RATED TEMPERATURE	INSULATION THICKNESS
Sharp freeze	-45º to -25º	11”
	-25º to -15º	10”
Freezer	-15º to -5º	8”
	- 5º to -20º	7”
Chiller	-5º to -20º	6”
	-20º to -32º	5”
Cooler	-32º to -45º	5”
	-45º to -60º	3”

INSULATION
(Cost per square foot)

Costs should be applied to the actual insulated surface area: walls (no adjustment), floors (deduct 35%), and ceilings (add 35%). Costs are for installation including vapor; barriers, fasteners, furring, framing, finishes, sealants, etc., as appropriate for each type of insulation. See Section SEG 4 for prefabricated insulated sandwich wall and/or roof panels.

Insulation Thickness	Corkboard	Fiberglass Board	Foamglas Board	Isocyanurate Board	Mineral Wool Batts	Styrene Board	Urethane Board	Urethane Spray-on
1"	11.00	9.68	11.20	9.53	8.69	8.69	9.68	6.26
2"	12.60	9.83	13.20	11.20	8.99	9.08	11.35	8.69
4"	16.05	11.05	17.10	13.90	9.83	9.83	14.35	13.50
6"	19.75	11.50	21.05	17.10	10.65	11.15	17.60	18.65
8"	23.20	12.35	25.00	20.05	11.50	11.85	20.90	21.35
10"	26.75	13.20	29.00	-----	12.45	12.65	-----	-----
12"	30.00	14.10	33.00	-----	13.40	13.90	-----	-----

PLUS COLD STORAGE DOORS
(Cost per square foot)

Cold storage doors are priced by the outside surface area of each door. Costs are based on completely installed, hinged infitting doors, including hardware and gaskets. For other doors use the surface area with the adjustments given below.

Thickness	To 15 Sq. Ft.	20 Sq. Ft.	33 Sq. Ft.	>40 Sq. Ft.
2"	205.00	175.00	147.00	125.00
4"	221.00	197.00	159.00	142.00
6"	237.00	214.00	183.00	159.00
8"	252.00	230.00	205.00	175.00

DOOR COST ADJUSTMENTS

Stainless steel: add 50%
Vestibule doors: add 60%

Track doors: add 10% to 20%
Wood-clad doors:
  deduct 10% per side
Sliding doors, single: add 25%
  double: add 40% to 50%
Electric door operator,
  sliding doors:
  single, add $9,300;
  double, add $10,700
  hydraulic operation,
    add 15% to electric

Antifreeze heating cables:
add $57.00 per linear foot of cable

Pneumatic door operator, swinging door,
  treadle-activated:
  add $4,450 – $5,950 each
  double doors:
    add $6,350 – $8,050 pair
  electric eye activation: add 10% – 12%
  sliding doors:
  add 30% – 50% to operator
Observation window: add $915 – $1,200
Hardware: locking, $398 – $1,130,
  panic, add $1,060 – $2,110

PLUS REFRIGERATION
(Cost per cubic foot)

Refrigeration equipment costs are based on the interior volume of the rooms which are cooled. Costs are for complete normal installation including wiring, switch panels, starters, piping, and ancillary items necessary to maintain proper temperature levels based on typical insulation. Room loads with much in and out traffic, highly controlled atmosphere, etc., will vary and costs may vary plus or minus 20%.

Room Size	Sharp Freezer	Freezer	Chiller	Cooler
5,000 cu. ft.	3.99	3.34	2.78	2.36
25,000	3.45	2.84	2.37	1.94
50,000	3.26	2.68	2.24	1.81
100,000	3.08	2.52	2.05	1.68
200,000	2.86	2.37	1.93	1.56
300,000	2.77	2.28	1.83	1.50
500,000	2.65	2.12	1.74	1.43
1,000,000	-----	1.99	1.64	1.30
2,500,000	-----	1.86	1.49	1.18
5,000,000 & up	-----	1.72	1.41	1.09

Subsoil heating will cost $2.53 to $4.24 per square foot of floor area. Adjust for other sitework refinements from Segregated or Unit-in-Place cost sections.

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COLD STORAGE
SECTION UIP 7

WALK-IN BOXES

Large walk-in boxes containing from 50 to 500 square feet by 7 1/2 feet high, with floors. Costs are for prefabricated, galvanized smooth or embossed metal-clad insulated cooler units, including refrigeration equipment and one door. Equipment quality and ancillary items can vary and costs will vary a plus 15% to a minus 10%. For outside installations, add $13.55 to $27.00 per square foot of floor area.

	SQUARE FEET						
Temperature	50	100	150	200	300	400	500
32º to 60ºF base	19400.00	27800.00	34000.00	39500.00	48800.00	56250.00	63000.00
5º to 31ºF add	1740.00	2020.00	2360.00	2525.00	2750.00	6050.00	6050.00
-15º to 5ºF add	2750.00	3500.00	3950.00	10700.00	10700.00	10700.00	10700.00
-45º to 15ºF add	3250.00	4125.00	4725.00	12800.00	12800.00	12800.00	12800.00

For wood exterior and interior, deduct 5% each; aluminum, add 7% each; stainless steel, add 20% to 30% each; for stainless floors, add 15%. For each additional foot of height from base, add 6%. Without floor but with screeds, deduct 17%. Where walls of the building form exterior wall of the box, deduct $159 per linear foot. Partitions, including door, cost $398 to $720 per linear foot.

Air curtains, see Section UIP 3. Traffic and stripdoors, see Section UIP 5. Extra doors see Cold Storage Doors. For glass reach-in doors, use 2" doors less 10%; heated doors, add 15% to 35%. Doorway ramps, exterior, cost $595 to $1240 each; for interior, add 100%; stainless steel, add 35%. Shelving costs $38.75 to $80.50 per linear foot of shelf. For cantilevered, add 50%.

Small display type commercial boxes or boxes with reach in glass fronts should be priced from Section UIP 15.

Subsoil heating will cost $2.53 to $2.91 per square foot of floor area. Adjust for other sitework refinements from Segregated or Unit-in-Place cost sections.

ELEVATORS – ESCALATORS

GENERAL INFORMATION

1 Selective-Collective controlled elevators are the type usually found in small office buildings, hospitals and apartments. The car or cars will respond automatically to the pressure of a car or corridor button. The control system will collect and save signals and respond to them in order. They are normally used in single elevator installations and up to groups or banks of three.

2 Selectomatic-Automatic, Autotronic, Measured Demand and similar terms are used by various manufacturers as descriptive names for the type of controls on fully automatic elevators in medium and high speed operation. These controls not only collect signals and give responses to banks of three or more elevators, but distribute service among them on a balancing basis. The elevators may have auxiliary controls to allow for manual operation to reassure passengers, to allow for guides or to prevent misuse.

3 Express or penthouse elevators do not have openings at each floor. For each bypassed floor without an opening, add the Express Elevator cost per floor or stop.

4 Attended passenger elevators require full-time operator control. The high cost of operation of this type has made most of these elevators obsolete. Because they are no longer produced, the appropriate passenger operated costs can be applied, deducting 10% from the costs for manual controls.

5 Hydraulic elevators are moved by one or two hydraulic plungers under the car. They are suitable for low-speed, low-rise operation and are somewhat lower in cost than electric elevators in these applications, up to 7 floors.

6 Variable-voltage equipment includes a D.C. generator for each elevator. Varying the voltage provides a wide range of speeds and acceleration. Automatic leveling is usually included with this equipment.

7 A.C. rheostatic control equipment is suitable for low-speed, low-rise application.

8 Miscellaneous types include escalators, moving walks, vertical and inclined residential and wheelchair lifts, sidewalk elevators, dumbwaiters and open personnel lifts.

INSTRUCTIONS

Elevators and escalator costs are lump sums to be added to the Segregated Cost estimates or as refinements to the Calculator Method costs. In order to compute the costs, information on the number of elevators, type of control, capacity of cars, rated speeds and number of stops will usually be required. The number of elevators, their associated stops and the type of cabs will all influence the cost rank chosen.

The cost schedules in this section are based on average installations within each class. In large, newer installations it is advisable to apply to the elevator installation company or manufacturer for actual prices, particularly where custom cabs are encountered.

EXAMPLES

EXAMPLE 1 – (Passenger Elevator)

A medical building, three stories and basement, two automatic elevators, geared, selective-collective, variable voltage control, 200 feet per minute with 2,500-pound capacity.

BASE COST	180,000
4 stops @ $9,700	38,800
Cost per elevator	218,800
2 elevators	x 2
Total cost of elevators	437,600

EXAMPLE 2 – (Freight Elevator)

A warehouse, four stories, one freight elevator, speed 100 feet per minute, hydraulic operation with 8,000-pound capacity, push-button controls, front and rear openings on first floor and power operation on all doors.

Base Cost	81,250
4 stops @ $27,300	109,200
1 rear-door opening (power door)	26,500
Total cost of elevators	216,950

MISCELLANEOUS ACCESSIBILITY EQUIPMENT

(Cost each, except where noted)

	COST RANGE
Emergency evacuation, portable lifts, wheelchair,	
motorized climber	14800.00 – 17500.00
power scooters	9000.00 – 10300.00
chair seat, mechanical track stair climber	4375.00 – 6200.00
seat carriers, hand held	201.00 – 259.00
Portable ramps, including railing, per linear foot	284.00 – 364.00
platforms, per square foot	115.00 – 141.00

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ELEVATORS
SECTION UIP 7

PASSENGER ELEVATORS – SELECTIVE-COLLECTIVE

(passenger-operated geared electric and hydraulic elevators)

ELECTRIC, VARIABLE VOLTAGE CONTROL

SPEED (FEET/MINUTE)	CAPACITY (POUNDS)					
	1500	2000	2500	3000	4000	5000
100	$113,000 – $122,000	$130,000 – $145,000	$145,000 – $162,000	$159,000 – $181,000	$183,000 – $214,000	$205,000 – $247,000
150	130,000 – 144,000	149,000 – 167,000	164,000 – 189,000	180,000 – 206,000	205,000 – 244,000	228,000 – 275,000
200	144,000 – 161,000	162,000 – 187,000	180,000 – 208,000	197,000 – 230,000	222,000 – 264,000	245,000 – 295,000
250	155,000 – 178,000	176,000 – 203,000	194,000 – 225,000	208,000 – 247,000	234,000 – 283,000	261,000 – 312,000
300	166,000 – 192,000	187,000 – 216,000	203,000 – 240,000	222,000 – 262,000	248,000 – 297,000	272,000 – 328,000
350	175,000 – 203,000	197,000 – 231,000	214,000 – 253,000	231,000 – 275,000	259,000 – 308,000	283,000 – 342,000
400	183,000 – 214,000	205,000 – 244,000	225,000 – 265,000	240,000 – 287,000	267,000 – 322,000	294,000 – 351,000
PLUS COST/STOP	$9,600 – $10,000	$9,700 – $10,200	$9,700 – $10,200	$10,000 – $10,300	$10,200 – $10,400	$10,300 – $10,500

HYDRAULIC

SPEED (FEET/MINUTE)	CAPACITY (POUNDS)					
	1500	2000	2500	3000	4000	5000
50	$42,300 – $53,000	$51,750 – $65,250	$60,000 – $74,250	$68,000 – $85,000	$82,500 – $103,000	$96,000 – $120,000
75	53,000 – 65,250	63,250 – 77,750	73,000 – 89,500	81,250 – 100,000	97,250 – 122,000	111,000 – 140,000
100	62,750 – 73,750	73,000 – 88,500	86,250 – 102,000	93,250 – 112,000	110,000 – 135,000	125,000 – 155,000
125	70,500 – 82,500	82,500 – 98,000	94,000 – 111,000	102,000 – 125,000	120,000 – 148,000	135,000 – 169,000
150	77,750 – 89,500	90,250 – 107,000	102,000 – 122,000	112,000 – 135,000	130,000 – 159,000	145,000 – 181,000
200	90,250 – 103,000	105,000 – 122,000	116,000 – 137,000	127,000 – 152,000	145,000 – 178,000	162,000 – 201,000
PLUS COST/STOP	$17,500 – $18,400	$18,300 – $19,500	$19,500 – $20,600	$20,300 – $21,600	$21,100 – $23,100	$22,800 – $24,200

ELECTRIC, A.C. RHEOSTATIC CONTROL

SPEED (FEET/MINUTE)	CAPACITY (POUNDS)		
	1200	1500	2000
50	$69,750 – $83,250	$77,250 – $92,500	$86,750 – $106,000
100	93,000 – 107,000	100,000 – 118,000	112,000 – 132,000
150	110,000 – 125,000	117,000 – 135,000	129,000 – 149,000
PLUS COST/STOP	$9,150 – $9,550	$9,350 – $9,600	$9,600 – $10,000

SMALL ELEVATORS

Small office and apartment elevators with simple call system and push button control, four passenger cab, and two or three stops, cost $86,250 to $118,000.

OBSERVATION ELEVATORS

For glass observation cars, add $14,200 to $28,100 plus $1,760 per stop. Exterior installations will cost an additional $26,500 to $33,700. Custom designed cars can run as much as five to ten times the cost of standardized cabs.

EXPRESS ELEVATORS

Cost per bypassed floor, without an opening, is $3,375 to $4,500 per floor. With openings, use cost per stop from tables.

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ELEVATORS

SECTION UIP 7

PASSENGER ELEVATORS – SELECTOMATIC-AUTOMATIC

(completely automatic, gearless machines, group-controlled elevators)

SPEED (FEET/MINUTE)	CAPACITY (POUNDS)		
	2000	2500	3000
300	$259,000 – $287,000	$275,000 – $303,000	$290,000 – $320,000
400	286,000 – 315,000	301,000 – 334,000	320,000 – 353,000
500	314,000 – 348,000	334,000 – 370,000	351,000 – 389,000
600	345,000 – 386,000	370,000 – 408,000	389,000 – 431,000
700	384,000 – 428,000	406,000 – 450,000	431,000 – 476,000
800	426,000 – 472,000	440,000 – 498,000	475,000 – 523,000
1,000	515,000 – 575,000	545,000 – 604,000	576,000 – 637,000
1,200	628,000 – 700,000	665,000 – 739,000	703,000 – 778,000
1,400	768,000 – 854,000	812,000 – 900,000	860,000 – 951,000
PLUS COST/STOP	$9,700 – $10,400	$10,200 – $10,600	$10,400 – $11,200

SPEED (FEET/MINUTE)	CAPACITY (POUNDS)		
	3500	4000	5000
300	$306,000 – $339,000	$322,000 – $358,000	$361,000 – $400,000
400	337,000 – 375,000	354,000 – 395,000	398,000 – 440,000
500	373,000 – 411,000	394,000 – 437,000	440,000 – 486,000
600	411,000 – 454,000	434,000 – 481,000	486,000 – 534,000
700	453,000 – 501,000	479,000 – 529,000	536,000 – 587,000
800	501,000 – 551,000	529,000 – 584,000	590,000 – 650,000
1,000	612,000 – 673,000	645,000 – 712,000	723,000 – 792,000
1,200	745,000 – 821,000	789,000 – 868,000	882,000 – 964,000
1,400	910,000 – 1,001,000	962,000 – 1,056,000	1,079,000 – 1,173,000
PLUS COST/STOP	$10,600 – $11,400	$11,200 – $11,700	$11,600 – $12,100

EXPRESS ELEVATORS: Cost per bypassed floor, without an opening, is $3,900 – $5,300 per floor. With openings, use cost per stop from table.

ESCALATORS

Costs are averages per each moving stairway.

32" WIDTH 5,000 PERSONS PER HOUR		48" WIDTH 8,000 PERSONS PER HOUR	
RISE	COST	RISE	COST
10'	239000.00	10'	259000.00
12'	247000.00	12'	265000.00
14'	253000.00	14'	276000.00
18'	265000.00	18'	295000.00
22'	283000.00	22'	314000.00
25'	295000.00	25'	330000.00

For glass balustrade panels or stainless steel sides add $1,000 to $3,150 per foot of rise per unit.

OBSERVATION ELEVATORS: Price with cost additives listed on Page 3.

MOVING WALKS

Costs are averages per section, up to 2% gradient.

LENGTH	COST PER LINEAR FOOT			
Linear Feet	24" WIDE	36" WIDE	48" WIDE	54" WIDE
40	7800.00	8500.00	9450.00	9750.00
60	6500.00	6850.00	7300.00	7450.00
100	4375.00	4775.00	5250.00	5350.00
300	3175.00	3475.00	3850.00	3900.00
500	2875.00	3150.00	3375.00	3625.00
750	2600.00	2975.00	3100.00	3325.00
1,000	2525.00	2750.00	2975.00	3100.00
1,400	2250.00	2575.00	2800.00	2875.00
1,800	2060.00	2440.00	2575.00	2750.00

For gradients up to 20%, add 1% for each percent over two. Costs include handrails.

VERTICAL WHEELCHAIR PORCH LIFT: For 400# capacity with 5' maximum lift, cost is $15,200 to $21,600. For each additional foot of height to a maximum of twelve feet, add $2,750 to $4,175 per foot.

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ELEVATORS
SECTION UIP 7

PASSENGER ELEVATORS – SELECTOMATIC-AUTOMATIC

MISCELLANEOUS ELEVATORS

RESIDENTIAL ELEVATORS: The small handicapped or two or three passenger elevators found in single family dwellings cost $28,600 to $57,500 for two stops plus $3,900 to $6,150 for each additional stop. For larger capacities over 700 lbs. cost $60,500 to $85,000 for two stops plus $14,700 to $19,400 for each additional stop. For custom cabs (decorative hardwood, brass and glass, etc.), add 75% to 150%.

INCLINED RAILWAYS: Inclined elevators (chairlift) cost $11,600 to $17,300 for normal 14' to 17' run. Add $169.00 per foot for longer runs. Add $2,480 to $3,175 for a two passenger lift. Add $3,475 to $5,000 for each turn. Add 50% for wheelchair capability. For exterior (hillside) installation, add $2,800.

SIDEWALK ELEVATORS: With sidewalk doors, 2,000# to 3,000# capacity, 25 square foot platform, the cost is $61,750 to $89,500.

PERSONNEL LIFTS: Revolving vertical belts with one person platforms cost $25,900 to $32,500 plus $6,900 per stop over two.

WINDOW-WASHING LIFTS: Exterior building maintenance platforms, self-powered, 24' to 26', cost $86,750 to $124,000. Custom engineered platforms can run as much as two to four times the cost of standard lifts. Supporting davits cost $14,200 to $18,400 per pair and sockets $865 to $1,170 each.

DUMBWAITERS: Automatic electric dumbwaiters, 500# capacity, 50 FPM, stainless steel cab, cost $25,100 (manual doors) to $64,750 (power doors) plus $5,650 to $7,250 per stop over two. For 100-FPM operation, add 30%; 150 FPM, add 50%. For 200# capacity, deduct 25%; 75#, deduct 50%. For hand operation, deduct 50%.

FREIGHT ELEVATORS
(push-button operation)

HYDRAULIC

CAPACITY (POUNDS)	SPEED (FEET/MINUTE)				PLUS COST PER STOP	
	50	100	125	150	Manual Doors	Power Doors
2,000	$43,400 – $47,500	$57,500 – $63,250	$65,750 – $73,000	$74,250 – $84,250	$11,900 – $12,400	$22,600 – $23,100
3,000	50,000 – 54,750	64,000 – 67,250	71,250 – 79,250	81,250 – 90,250	12,600 – 13,100	23,700 – 24,400
4,000	53,750 – 60,500	68,000 – 75,500	76,500 – 85,000	86,000 – 96,000	13,400 – 13,900	24,400 – 25,300
5,000	59,500 – 65,250	72,500 – 80,000	80,000 – 88,500	88,500 – 100,000	13,900 – 14,800	25,300 – 26,500
6,000	64,000 – 69,000	75,500 – 84,250	83,250 – 93,000	93,000 – 103,000	14,800 – 15,300	26,200 – 27,200
8,000	69,750 – 75,500	81,250 – 91,000	86,000 – 98,000	97,250 – 109,000	15,600 – 16,200	27,300 – 28,400
10,000	104,000 – 116,000	123,000 – 135,000	133,000 – 147,000	144,000 – 159,000	16,200 – 16,700	28,100 – 29,400
12,000	123,000 – 136,000	144,000 – 159,000	154,000 – 173,000	167,000 – 189,000	16,700 – 17,600	29,000 – 29,800
15,000	150,000 – 167,000	175,000 – 195,000	187,000 – 209,000	200,000 – 226,000	18,100 – 18,400	29,700 – 31,100
20,000	195,000 – 216,000	223,000 – 251,000	239,000 – 272,000	261,000 – 294,000	19,100 – 19,700	30,800 – 32,200

ELECTRIC, VARIABLE VOLTAGE CONTROL

CAPACITY (POUNDS)	SPEED (FEET/MINUTE)				PLUS COST PER STOP	
	100	200	300	400	Manual Doors	Power Doors
2,500	$130,000 – $144,000	$146,000 – $162,000	$166,000 – $184,000	$189,000 – $208,000	$12,400 – $13,000	$23,400 – $23,900
4,000	138,000 – 154,000	159,000 – 178,000	184,000 – 203,000	211,000 – 234,000	13,400 – 14,700	24,500 – 25,800
5,000	143,000 – 158,000	166,000 – 184,000	194,000 – 214,000	225,000 – 250,000	14,700 – 15,100	25,800 – 26,700
6,000	146,000 – 162,000	170,000 – 191,000	201,000 – 223,000	237,000 – 262,000	15,100 – 15,600	26,500 – 27,300
8,000	152,000 – 169,000	181,000 – 200,000	214,000 – 239,000	256,000 – 281,000	15,900 – 16,200	27,800 – 28,600
10,000	162,000 – 181,000	197,000 – 216,000	234,000 – 262,000	284,000 – 314,000	16,600 – 17,300	28,600 – 29,500
15,000	206,000 – 228,000	275,000 – 303,000	----- – -----	----- – -----	18,100 – 18,400	30,300 – 31,100
20,000	244,000 – 272,000	----- – -----	----- – -----	----- – -----	19,100 – 19,700	31,200 – 32,500

REAR DOORS: For rear-door openings, add $15,000 to $15,900 for the first rear door, if manually operated, plus $11,600 to $12,600 for each additional manual door. For power operated rear doors, use $26,500 to $28,000 for the first door and $22,500 to $24,200 for each additional door.

SELECTIVE-COLLECTIVE OPERATION: Add 10%.

A.C. RHEOSTATIC: Use 150% to 175% of the base cost of a comparable hydraulic elevator plus 50% to 75% of the cost per stop.

MANUAL CONTROLS: Deduct 10% from base cost and use manual door cost for stops.

PARKING LIFTS

Average costs per single elevated platform stall for four column surface mounted automobile storage lifts. For additional raised stall, add $2,875 to $5,700 each.

	COST RANGE
3,000#, electric winch	14300.00 – 15600.00
4,000#, hydraulic	17500.00 – 19400.00
7,000#, hydraulic	20000.00 – 23300.00

PNEUMATIC TUBE SYSTEMS

Cost per station, complete. For simple two-station systems, deduct 50% to 75%.

3" or 4"	30500.00 – 40100.00
6" or 4" x 7"	43900.00 – 53500.00

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CONVEYING AND MATERIAL HANDLING SYSTEMS

SECTION UIP 7

GENERAL INFORMATION

Material handling systems can have a wide range in costs depending on the operational loads and conditions placed on the equipment. The costs listed below represent averages of standard (medium) applications. For heavy or severe service requirements, such as continuous and repetitive operations at rated capacity or with heavy attachments or operations under extreme atmospheric conditions, the costs may run 100% above those listed, whereas very light applications (e.g. single girder bridge) can run 25% to 50% below the listed costs. For outside operations, add 10%. For multiple hoists, use the total lifting capacity of all hoists to determine crane or craneway capacity, always selecting the next higher size.

BRIDGE CRANES

Costs are averages for ground controlled, variable speed, twingirder, overhead cranes, exclusive of craneways. For cranes with cabs, add $9,550 for minimum controls to $34,700 for deluxe cabs with air conditioning and complete controls.

CAPACITY (TONS)	SPAN					
	20'	30'	40'	50'	75'	100'
2	122000.00	135000.00	149000.00	166000.00	214000.00	280000.00
3	125000.00	138000.00	153000.00	169000.00	217000.00	284000.00
5	133000.00	146000.00	161000.00	180000.00	228000.00	295000.00
7-1/2	142000.00	156000.00	173000.00	189000.00	242000.00	306000.00
10	153000.00	167000.00	183000.00	201000.00	253000.00	320000.00
15	176000.00	192000.00	208000.00	226000.00	281000.00	348000.00
20	201000.00	217000.00	239000.00	258000.00	312000.00	379000.00
25	234000.00	251000.00	272000.00	290000.00	347000.00	412000.00

JIB CRANES

Costs are averages for self supporting jib or pillar base cranes with a 360° rotation, completely installed including mast, foundation and miscellaneous fittings. For each foot of variation from an underbeam base height of 10 feet, add or deduct $74.50 to $226. For boom electrification, add $1,410. For 200° knee brace, wall bracket or column mount installations, deduct 50% to 60%.

BOOM LENGTH (Feet)	CAPACITY					
	1/4	1/2	1	2	3	4
8	3375.00	3725.00	4400.00	6150.00	7650.00	9200.00
12	3900.00	4500.00	5700.00	7900.00	10100.00	12400.00
16	4850.00	5700.00	7250.00	10200.00	13400.00	16200.00
20	5850.00	7050.00	9150.00	13400.00	17600.00	21600.00

CRANEWAY SPANS

Crane runways, bracketed or braced to the buildings' column framing members, cost from $226 to $493 per linear foot for each rail. Light monorail trolley runways cost from $24.30 to $178 per linear foot. For free standing, self-supporting runways, add 50% to 80%. For curved runways, add 75% to 150%. Costs may vary plus or minus 10% depending on overall length and height of installation. For detailed crane or trolleyways, use steel beam costs in Section UIP 1 plus 10% to 20%. For electrical conductor assembly add $34.00 to $68.50 per linear foot. Curved conductor, add 25% to 35%. Examples including ±10% range:

BRIDGE CRANEWAY		TROLLEYWAY	
Capacity	Cost Range/Lin. Ft.	Capacity	Cost Range/Lin. Ft.
2 ton	226.00 - 284.00	1 ton	24.30 - 28.00
10	348.00 - 381.00	2	38.75 - 52.50
25	404.00 - 493.00	7-1/2	154.00 - 178.00

GANTRY CRANES

Costs are averages of portable steel gantry cranes with a maximum adjustable height of 15 feet. For each foot of deviation from the 15-foot base, add or deduct $194 to $525 per foot. For fixed height gantries, deduct 25%. For all-aluminum construction, add 50% to 100%. Gantries with adjustable spans, add $2,170 to $3,600. Angle iron tracks cost $24.75 to $43.25 per linear foot. For power operation, add $13,800 to $21,600.

SPAN (Feet)	CAPACITY (TONS)					
	1	2	3	5	7-1/2	10
10	5250.00	7400.00	9150.00	11700.00	14300.00	16400.00
15	6050.00	8500.00	10500.00	13900.00	16700.00	19500.00
20	6850.00	9750.00	12200.00	15800.00	19500.00	22800.00
30	8450.00	12300.00	15100.00	20000.00	24800.00	29400.00

HOISTS AND TROLLEYS

Costs are averages of hoists with a maximum lift of 10 feet. For each foot over 10 feet, add $42.25 to $178.00 per foot to the costs and add 50% to 70% to the base costs over 15 feet. Motors are single speed. Add 10% to 30% each for two speed or variable speed or direct current operation. For high-speed trolleys (over 75 feet per minute), add 200%. For integral trolley hoists, add 15% to 25%

HOISTS

CAPACITY (Tons)	MANUAL COST RANGE	ELECTRIC COST RANGE
1/2	315.00 - 1450.00	2675.00 - 3400.00
1	437.00 - 1700.00	3050.00 - 4100.00
2	685.00 - 2450.00	3850.00 - 5800.00
3	900.00 - 3125.00	4575.00 - 7350.00
5	1340.00 - 4450.00	6300.00 - 10400.00
7-1/2	2050.00 - 6200.00	8100.00 - 14700.00
10	2625.00 - 7750.00	10100.00 - 18400.00
15	---- - ----	14100.00 - 26500.00
20	---- - ----	17800.00 - 34400.00
30	---- - ----	25900.00 - 50250.00

TROLLEYS

CAPACITY (TONS)	PLAIN (FREE WHEELING) COST RANGE	GEARED COST RANGE	MOTORIZED COST RANGE
1/2	217.00 - 660.00	315.00 - 1450.00	2450.00 - 5550.00
1	292.00 - 805.00	437.00 - 1620.00	2575.00 - 5750.00
2	486.00 - 1160.00	685.00 - 2120.00	2750.00 - 6150.00
3	620.00 - 1490.00	865.00 - 2480.00	2975.00 - 6500.00
5	865.00 - 2300.00	1370.00 - 3375.00	3375.00 - 7100.00
7-1/2	1290.00 - 3100.00	2060.00 - 4375.00	3800.00 - 7950.00
10	1760.00 - 3900.00	2600.00 - 5450.00	4250.00 - 9000.00
15	2575.00 - 5750.00	3850.00 - 7550.00	5150.00 - 10700.00
20	---- - ----	---- - ----	6200.00 - 12800.00
30	---- - ----	---- - ----	8100.00 - 16200.00

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CONVEYING AND MATERIAL HANDLING SYSTEMS

SECTION UIP 7

GRAVITY ROLLER CONVEYORS

Costs are average costs per section for steel conveyor sections with galvanized steel rollers and include supports and installation. Light duty conveyor frame capacity is 900 pounds with H-stand supports on 10' centers and 1,300 pounds with H-stand supports on 5' centers. Each 10' section consists of 40 2" OD rollers set on 3" centers. Each roller has a 50-pound load capacity.

The medium duty frame capacity is 1,400 pounds with H-stand supports on 10' centers and 3,200 pounds with H-stand supports on 5' centers. Each 10' section consists of 40 2" OD rollers set on 3" centers. Each roller has a 250-pound load capacity.

The heavy duty frame capacity is 2,500 pounds with H-stand supports on 10' centers and 7,600 pounds with H-stand supports on 5' centers. Each 10' section consists of 30 2-½" OD steel rollers set on 4" centers. Each roller has a 750-pound load capacity.

TYPE	OVER-ALL WIDTH	BETWEEN RAIL WIDTH	STRAIGHT SECTION		CURVED	
			10'	5'	90°	45°
LIGHT DUTY	12"	10"	945.00	620.00	660.00	493.00
	18"	16"	1130.00	710.00	710.00	510.00
	20"	18"	1160.00	735.00	735.00	515.00
	24"	22"	1280.00	780.00	770.00	525.00
MEDIUM DUTY	12"	10"	1120.00	695.00	1210.00	875.00
	18"	16"	1280.00	780.00	1360.00	975.00
	24"	20"	1410.00	865.00	1410.00	1020.00
	30"	28"	1580.00	935.00	1520.00	1110.00
HEAVY DUTY	12"	10"	2340.00	1460.00	1410.00	1020.00
	18"	16"	2725.00	1620.00	1590.00	1160.00
	30"	20"	3425.00	2010.00	1690.00	1360.00
	42"	28"	4100.00	2390.00	2160.00	1560.00

MICROROLLER CONVEYORS

Costs are average costs per section for steel conveyor sections with 3/4" OD galvanized steel rollers with a 30-pound load capacity for each roller and a 350-pound load capacity per section. Supports and installation costs are included.

Aluminum conveyor sections with aluminum rollers of the same size have 40% less capacity and cost 15% more.

ROLLER CENTERS	OVERALL WIDTH	BETWEEN RAIL WIDTH	STRAIGHT SECTION	
			10 FEET	5 FEET
1"	6"	5"	1550.00	920.00
	12"	10-1/2"	1800.00	1030.00
	15"	14"	1870.00	1110.00
	19"	18"	2000.00	1150.00
2"	6"	5"	935.00	635.00
	12"	10-1/2"	1020.00	685.00
	15"	14"	1070.00	710.00
	19"	18"	1150.00	735.00

SKATE WHEEL CONVEYORS

Skate wheel conveyors are used to move lightweight cartons and packages from one station to another, in warehouses and assembly areas. Costs are average costs per section for steel conveyors with 2 x 5/8" ball bearing wheels on 1/4" axles at 3" centers. 10' section frame capacity is 900 pounds. Each wheel load capacity is 60 pounds.

Costs are average cost per section for flexible expandable skate wheel conveyors, including supports. For cost of flexible roller conveyors add 18%.

OVERALL WIDTH	WHEELS PER FT.	STRAIGHT SECTION		CURVED	
		10 FT.	5FT.	90°	45°
12"	6	620.00	429.00	----	----
	10	720.00	493.00	670.00	445.00
	12	770.00	510.00	----	----
	16	865.00	575.00	----	----
15"	8	695.00	493.00	----	----
	10	735.00	510.00	----	----
	14	865.00	575.00	----	----
	16	915.00	595.00	----	----
18"	12	835.00	540.00	----	----
	14	865.00	580.00	----	----
	16	920.00	620.00	----	----
	18	1000.00	635.00	----	----
20"	16	1010.00	670.00	----	----
	18	1070.00	695.00	----	----
	20	1130.00	710.00	875.00	595.00
	24	1240.00	770.00	----	----
24"	18	1140.00	720.00	----	----
	20	1170.00	750.00	1160.00	805.00
	24	1300.00	795.00	----	----
	28	1380.00	850.00	----	----

ACCORDION-TYPE FLEXIBLE CONVEYORS

Costs are average costs per section for flexible expandable skate wheel conveyors, including supports. For cost of flexible roller conveyors add 18%.

WIDTH	ADJUSTABLE LENGTH (FT.)	CAPACITY (LB./LIN. FT.)	WHEELS NYLON	PLASTIC	STEEL
14"	2' - 8'	58	1500.00	1590.00	1730.00
	3' - 12'	58	2725.00	2850.00	3175.00
	5' - 18'	58	3650.00	3975.00	4325.00
	6' - 24'	58	4600.00	5100.00	5600.00
18"	2' - 8'	90	1730.00	1830.00	2000.00
	3' - 12'	90	3200.00	3325.00	3875.00
	5' - 18'	90	3575.00	4250.00	4525.00
	6' - 24'	90	4850.00	5750.00	5950.00
24"	2' - 8'	120	2120.00	2310.00	2450.00
	3' - 12'	120	3900.00	4100.00	4400.00
	5' - 18'	120	5400.00	5950.00	6200.00
	6' - 24'	120	7550.00	8050.00	8450.00

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CONVEYING AND MATERIAL HANDLING SYSTEMS

SECTION UIP 7

BELT-DRIVEN LIVE ROLLER CONVEYORS

Belt-driven live roller conveyors are used for moving lightweight boxes and packages where product positioning is necessary. Costs are average costs per section for a 60-FPM conveyor powered by a 1/2-HP 220/3/60 motor, with 2” OD rollers and 6” PVC belts, including supports and installations.

OVERALL WIDTH	BETWEEN RAIL WIDTH	ROLLER CENTERS	OVERALL LENGTH 10’	15’	20’	25’	30’
12”	9”	3"	9550.00	10700.00	12100.00	13400.00	14700.00
		4-1/2"	8650.00	9750.00	10700.00	11700.00	12800.00
		6"	8450.00	10800.00	10300.00	11200.00	12100.00
		12"	8100.00	9000.00	9700.00	10400.00	11200.00
18”	15”	3"	10000.00	11400.00	12900.00	14200.00	15600.00
		4-1/2"	9350.00	10400.00	11600.00	12800.00	13900.00
		6"	9050.00	10100.00	11200.00	12100.00	13100.00
		12"	8500.00	9600.00	10300.00	11200.00	12000.00
24”	21”	3"	10700.00	12400.00	14100.00	15600.00	17300.00
		4-1/2"	10100.00	11300.00	12600.00	14000.00	15200.00
		6"	9750.00	10900.00	12100.00	13400.00	14500.00
		12"	9350.00	10300.00	11300.00	12300.00	13300.00
30”	27”	3"	11300.00	13200.00	15100.00	16700.00	18700.00
		4-1/2"	10700.00	12200.00	13700.00	15100.00	16600.00
		6"	10200.00	11400.00	12900.00	14000.00	15200.00
		12"	9750.00	10700.00	11900.00	12900.00	14000.00
36”	33”	3"	12000.00	14100.00	16100.00	18100.00	20300.00
		4-1/2"	11200.00	12900.00	14500.00	15900.00	17600.00
		6"	10600.00	12100.00	13600.00	14800.00	16200.00
		12"	10200.00	11300.00	12400.00	13600.00	14800.00
42”	39”	3"	12600.00	14800.00	16900.00	19400.00	21400.00
		4-1/2"	11900.00	13600.00	15200.00	16900.00	18900.00
		6"	11200.00	12800.00	14200.00	15600.00	17300.00
		12"	10600.00	12000.00	13200.00	14500.00	15600.00

CHAIN DRIVEN LIVE ROLLER CONVEYORS

Chain driven live roller conveyors are used for moving heavy pallets and drums. Costs are average for a 40-FPM conveyor powered by a 3/4-HP 220/3/60 motor, with 2-½” OD rollers at 6” centers, No. 50 chain, roller to roller drive, including supports and installations.

OVERALL WIDTH	BETWEEN RAIL WIDTH	OVERALL LENGTH 5’	10’	15’	20’	25’	30’
28"	24"	9600.00	10200.00	13100.00	15900.00	18900.00	21600.00
34"	30"	10200.00	10600.00	13700.00	16600.00	19400.00	22500.00
40"	36"	10300.00	10900.00	14100.00	16700.00	19700.00	22800.00
46"	42"	10700.00	11200.00	14700.00	18100.00	20900.00	24000.00

FLOOR TO FLOOR INCLINE CONVEYORS

Incline conveyors are used to move product from level to level or serve as a booster in a gravity system. Costs are average costs per section for a 60-FPM conveyor powered by a 1/2-HP 115/160 motor, with a rubber top belt and a 400-pound load capacity. Supports and installation are included.

OVERALL WIDTH	BELT WIDTH	FLOOR TO FLOOR LENGTH 8’	9’	10’	11’	12’	13’	14’	15’
		OVERALL LENGTH 21’	23’	25’	27’	29’	31’	33’	35’
15"	12"	9850.00	10200.00	10500.00	10800.00	11200.00	11600.00	12000.00	12300.00
22"	18"	11300.00	11900.00	12300.00	12900.00	13300.00	13900.00	14200.00	14800.00
28"	24"	12400.00	13000.00	13700.00	14200.00	14800.00	15400.00	15900.00	16600.00
34"	30"	14100.00	14800.00	15500.00	16200.00	16900.00	17800.00	18400.00	19200.00

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TANKS
SECTION UIP 8

DEFINITIONS AND COMMENTS

API refers to the standard specifications of the American Petroleum Institute.

ASME refers to the standard specifications for pressure tank design of the American Society of Mechanical Engineers.

WATER TANKS are normally measured in gallons.

OIL TANKS are normally measured in barrels of 42 gallons each.

STEEL RING CURB is a steel ring used to hold the foundation sand or gravel in place. The tank bottom is then placed on the sand. All tanks do not have curb rings.

HORTON SPHERE, HEMISPHEROID, DEWAR, etc., are types of large liquid- and gas-pressure vessels.

Many special tanks found in use cannot be included here, such as those for storage of exotic fuels, or those used in food or beverage processing which are in the nature of industrial processing equipment.

The costs of the tanks listed in this section are averages of total costs in place at the site including necessary foundations and tank fittings, but not pilings, pipe, fencing, site roads, etc.

The tanks included in this section are those used primarily for liquid and industrial storage.

The tank costs listed do not include an allowance for cathodic protection. Add 2% – 5% of the tank cost when found.

WEIGHTS AND MEASURES

1 GALLON (WATER)	WEIGHS	8.34 Pounds
1 gallon	equals	.1337 cubic foot
1 gallon	equals	.1074 bushel
1 gallon	equals	.8327 Imperial gallon
1 gallon	equals	3.7853 liters
1 acre foot	equals	325,900 gallons
1 cubic foot	equals	.8032 bushel
1 barrel (oil)	equals	42 gallons
1 barrel (water)	equals	31.5 gallons

Pressure in pounds per square inch of column of water equals .434 times the height of the column in feet.

Circumference of a Circle	=	3.1416 x the diameter
Area of a circle	=	.7854 x square of the diameter
Area of an ellipse	=	.7854 x product of both diameters
Volume of a sphere	=	.5236 x cube of the diameter
Volume of a cone	=	area of base x 1/3 of the altitude

Capacity in barrels (oil) = D^2 x .1399 x height
(diameter and height in feet)

Capacity in gallons = D^2 x 5.8748 x height
(diameter and height in feet)

Capacity in bushels = D^2 x .6308 x height
(diameter and height in feet)

CAPACITY OF CYLINDRICAL TANKS OR RESERVOIRS

(Per foot of depth or height.)

DIA. (feet)	U.S. BARRELS (gallons)	(42 gal.)	DIA. (feet)	U.S. BARRELS (gallons)	(42 gal.)
1	5.87	0.14	32	6,016.20	143.20
2	23.50	0.56	33	6,398.10	152.30
3	52.88	1.26	34	6,790.70	161.60
4	94.00	2.24	35	7,196.00	171.30
5	146.88	3.50	36	7,613.30	181.30
6	211.51	5.00	37	8,041.90	191.50
7	287.88	6.80	38	8,482.40	202.00
8	376.01	9.00	39	8,934.90	212.70
9	475.89	11.30	40	9,398.70	223.80
10	587.52	14.00	41	9,875.80	235.10
11	710.90	16.90	42	10,362.00	246.70
12	846.03	20.20	43	10,861.60	258.60
13	992.91	23.70	44	11,374.00	270.80
14	1,151.50	27.40	45	11,895.30	283.20
15	1,321.90	31.50	46	12,430.10	296.00
16	1,504.10	35.80	47	12,976.10	309.00
17	1,697.90	40.40	48	13,534.80	322.30
18	1,903.60	45.30	49	14,104.00	335.80
19	2,120.90	50.50	50	14,685.90	349.70
20	2,350.10	56.00	60	21,149.30	503.60
21	2,591.00	61.70	70	28,786.50	685.50
22	2,843.60	67.70	80	37,598.70	895.30
23	3,108.00	74.00	90	47,585.90	1,133.10
24	3,384.10	80.60	100	58,748.00	1,399.00
25	3,672.00	87.40	120	84,597.10	2,014.50
26	3,971.60	94.60	140	115,146.10	2,742.00
27	4,283.00	102.00	160	150,394.90	3,581.40
28	4,606.20	109.70	180	190,343.50	4,532.70
29	4,941.00	117.60	200	234,992.00	5,596.00
30	5,287.70	125.80	220	284,340.30	6,771.20
31	5,646.10	134.40	240	338,388.50	8,056.90

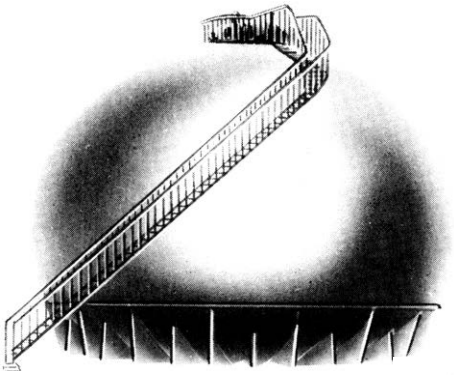
TYPICAL TANK LIVES

Tank lives can vary widely depending on the storage loads and conditions placed on the individual tank, the method of installation and appropriate maintenance and warranties. The typical lives listed below represent averages under standard applications. Lives may be shortened under severe requirements, such as extremely corrosive materials and/or atmospheric conditions or lengthened under very mild circumstances, by special coatings, double walls, etc.

	Years
Concrete tanks	30 – 55
Galvanized steel	15 – 20
Steel oil storage	25 – 30
Steel surface water storage	25 – 50
Elevated steel tanks	30 – 40
Underground steel, single wall	10 – 20
Double walled	25 – 35
Fiber coated	25 – 35
Galvanized steel chemical storage	3 – 17
Stainless steel chemical storage	15 – 30
Polyethylene chemical storage	15 – 20
Fiberglass chemical storage	15 – 20
Underground fiberglass	30 – 35
Wood	20 – 35
Steel pressure tanks	20 – 40

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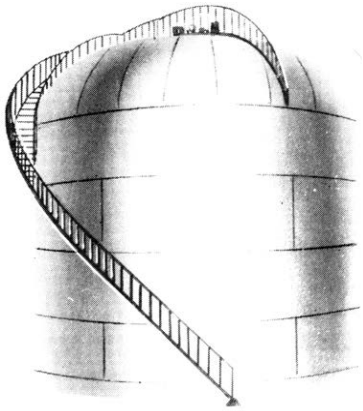
TANKS
SECTION UIP 8



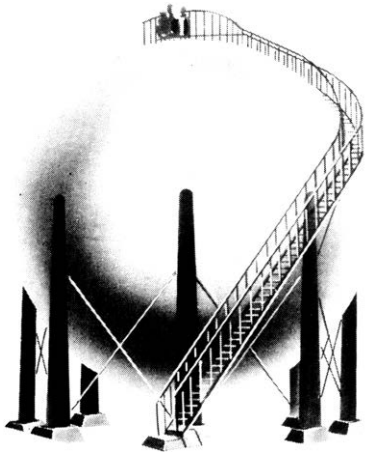
SPHEROID



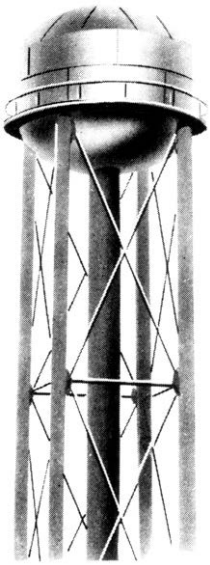
WOOD TANK ON TOWER



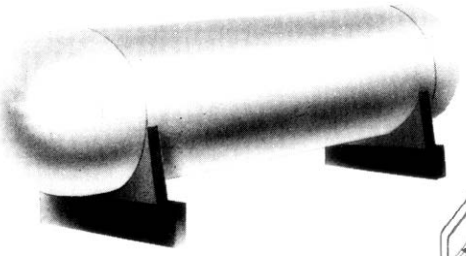
HEMISPHEROID



SPHERE FOR LIQUID STORAGE



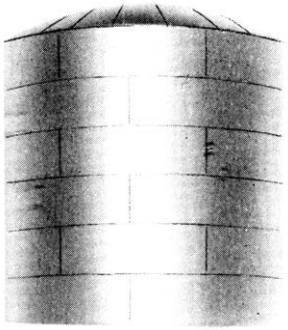
ELEVATED WATER TANK



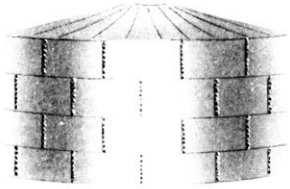
HORIZONTAL PRESSURE TANK



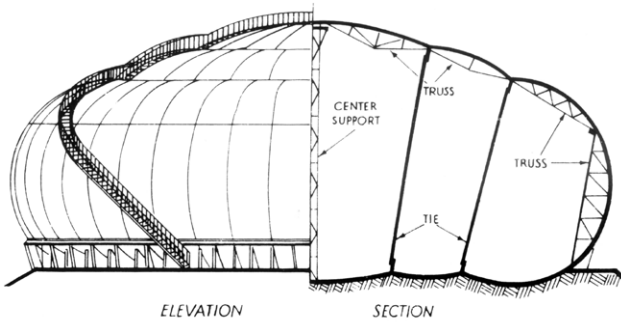
BOLTED STEEL – API



WELDED STEEL WATER TANK

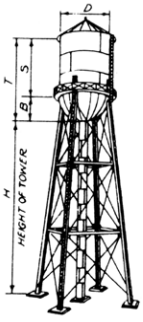


GALVANIZED BULK STORAGE



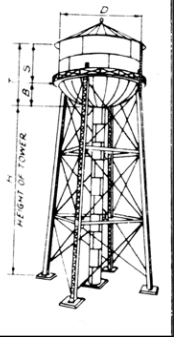
HORTON SPHEROID

HEMISPHERICAL BOTTOM TANKS



CAPACITY (GALLONS)	DIAM. D	SIDE S	DEPTH T
3,000	8'	5' – 5''	9' – 5''
5,000	8'	10' – 10''	14' – 10''
10,000	10'	13' – 9''	18' – 9''
15,000	12'	13' – 9''	19' – 9''
20,000	14'	12' – 9''	19' – 9''
25,000	14'	17' – 1''	24' – 1''
30,000	16'	14' – 9''	22' – 9''
35,000	16'	18' – 1''	26' – 1''
40,000	16'	21' – 7''	29' – 7''
50,000	18'	20' – 3''	29' – 3''
60,000	20'	18' – 11''	28' – 11''
75,000	20'	25' – 5''	35' – 5''
100,000	24'	21' – 7''	33' – 7''
125,000	24'	29' – 0''	41' – 0''
150,000	26'	29' – 2''	42' – 2''
175,000	28'	28' – 9''	42' – 9''
200,000	28'	34' – 2''	48' – 2''
250,000	32'	30' – 11''	46' – 11''
300,000	34'	32' – 11''	49' – 11''
500,000	40'	40' – 0''	60' – 0''

CAPACITY (GALLONS)	DIAM. D	SIDE S	TOTAL T
20,000	16'	10' – 9''	14' – 9''
25,000	18'	10' – 3''	14' – 9''
30,000	18'	12' – 11''	17' – 5''
35,000	20'	11' – 9''	16' – 9''
40,000	20'	13' – 11''	18' – 11''
50,000	22'	13' – 11''	19' – 5''
60,000	24'	13' – 11''	19' – 11''
75,000	26'	14' – 9''	21' – 3''
100,000	30'	13' – 11''	21' – 5''
125,000	30'	13' – 9''	26' – 3''
150,000	34'	16' – 7''	25' – 1''
175,000	36'	17' – 1''	26' – 1''
200,000	38'	17' – 5''	26' – 11''
250,000	40'	20' – 1''	30' – 1''
300,000	40'	25' – 5''	35' – 5''
400,000	47'	23' – 1''	34' – 10''
500,000	51'	24' – 5''	37' – 2''
600,000	54'	26' – 1''	39' – 7''
750,000	58'	28' – 5''	42' – 11''
1,000,000	66'	28' – 3''	44' – 9''



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TANKS
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ELEVATED STEEL TANKS

Costs are averages for each of the high stress and low stress areas. Costs include tank, tower, riser pipe, ladder, and other equipment normally installed, completely erected as well as typical foundations and painting.

LOW STRESS AREAS

Average costs in areas not requiring earthquake (including zone 1 areas) or hurricane resistant structures.

CAPACITY	TOWER HEIGHT			
(gallons)	50'	75'	100'	150'
25,000	536000.00	595000.00	690000.00	894000.00
50,000	574000.00	635000.00	728000.00	940000.00
75,000	687000.00	761000.00	851000.00	1059000.00
100,000	743000.00	813000.00	909000.00	1120000.00
150,000	909000.00	998000.00	1091000.00	1302000.00
200,000	1224000.00	1323000.00	1424000.00	1625000.00
300,000	1523000.00	1665000.00	1772000.00	1970000.00
400,000	1790000.00	1955000.00	2046000.00	2266000.00
500,000	1997000.00	2171000.00	2341000.00	2583000.00
750,000	2617000.00	2832000.00	3081000.00	3467000.00
1,000,000	3313000.00	3571000.00	3896000.00	4367000.00
1,500,000	4608000.00	4958000.00	5423000.00	6139000.00
2,000,000	5892000.00	6332000.00	6943000.00	7922000.00

HIGH-STRESS AREAS

Average costs in areas requiring earthquake (zones 3 and 4) or hurricane resistant structures.

CAPACITY	TOWER HEIGHT			
(gallons)	50'	75'	100'	150'
25,000	676000.00	754000.00	876000.00	1123000.00
50,000	731000.00	812000.00	923000.00	1189000.00
75,000	861000.00	958000.00	1085000.00	1340000.00
100,000	943000.00	1032000.00	1158000.00	1422000.00
150,000	1155000.00	1259000.00	1382000.00	1646000.00
200,000	1549000.00	1677000.00	1808000.00	2060000.00
300,000	1929000.00	2119000.00	2234000.00	2499000.00
400,000	2265000.00	2478000.00	2589000.00	2873000.00
500,000	2533000.00	2751000.00	2962000.00	3268000.00
750,000	3319000.00	3589000.00	3902000.00	4395000.00
1,000,000	4199000.00	4526000.00	4935000.00	5535000.00
1,500,000	5830000.00	6282000.00	6877000.00	7775000.00
2,000,000	7468000.00	8028000.00	8801000.00	10042000.00

RESERVOIRS

Average cost of cut and fill reservoirs with concrete or asphalt linings and metal, wood or concrete roof structures, per unit of rated capacity.

Roof Cover	Cost per Gallon	Cost per Acre Foot
Floating	$.39	$126,000
Steel	.81	262,000
Aluminum	.91	296,000
Concrete	.95	308,000

WELDED STEEL WATER TANKS

Costs are average costs of surface reservoirs including typical tank ancillaries such as roofs, ladders, painting, fittings on tank, etc. Steel standpipes (height exceeds diameter) will cost 35% more. Sand and gravel foundations with steel retaining rings are included on those of 1,000,000 gallons capacity, or less, concrete foundations on larger tanks.

CAPAC.	COST	CAPAC.	COST	CAPAC.	COST
(gallons)		(gallons)		(gallons)	
10,000	77000.00	200,000	461000.00	2,000,000	2132000.00
20,000	123000.00	250,000	520000.00	2,500,000	2495000.00
30,000	163000.00	300,000	575000.00	3,000,000	2852000.00
50,000	221000.00	400,000	720000.00	4,000,000	3478000.00
75,000	287000.00	500,000	845000.00	5,000,000	4071000.00
100,000	349000.00	750,000	1085000.00	6,000,000	4642000.00
125,000	375000.00	1,000,000	1254000.00	7,500,000	5420000.00
150,000	406000.00	1,500,000	1750000.00	10,000,000	6622000.00

BOLTED STEEL WATER TANKS

Costs are average costs for factory coated, bolted steel surface reservoirs erected on sand or gravel with a steel ring curb, including typical accessories such as roof, ladders, manways, vents, fittings on tank, and liquid level indicators, etc. Concrete foundations cost an additional $10.50 to $13.25 per cubic foot. Tank and foundation costs depend on seismic zone. For areas requiring earthquake resistant structures add 5% to the tank cost and use $11.30 per cubic foot for a concrete foundation cost. Modular knockdown containment tanks with membrane liner and cover will cost 5% to 15% less.

CAPAC.	DIMEN.	TANK	CAPAC.	DIMEN.	TANK
(gallons)	(d x h)	COST	(gallons)	(d x h)	COST
10,000	15' x 8'	46400.00	300,000	47' x 24'	331000.00
30,000	18' x 16'	83250.00	400,000	53' x 24'	418000.00
100,000	26' x 20'	135000.00	500,000	60' x 24'	528000.00
125,000	29' x 24'	165000.00	600,000	64' x 24'	615000.00
150,000	32' x 24'	195000.00	900,000	78' x 24'	864000.00
200,000	37' x 16'	243000.00	1,200,000	91' x 24'	1146000.00

CONCRETE WATER TANKS

Costs are averages of completely erected surface reservoirs, including foundations, dome roofs, and typical tank ancillaries. Sitework and exterior piping are extra. Small precast modular tanks, up to 30,000 gallons, cost $11,000 to $13,300 per 5,000 gallon module.

CAPAC.	COST	CAPAC.	COST	CAPAC.	COST
(gallons)		(gallons)		(gallons)	
10,000	88000.00	200,000	683000.00	2,000,000	2091000.00
20,000	139000.00	250,000	762000.00	2,500,000	2336000.00
30,000	182000.00	300,000	830000.00	3,000,000	2705000.00
50,000	291000.00	400,000	956000.00	4,000,000	3246000.00
75,000	392000.00	500,000	1067000.00	5,000,000	3811000.00
100,000	464000.00	750,000	1299000.00	6,000,000	4426000.00
125,000	523000.00	1,000,000	1493000.00	7,500,000	5164000.00
150,000	594000.00	1,500,000	1819000.00	10,000,000	6637000.00

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TANKS
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WOOD TANKS

Costs are averages for redwood or fir tanks completely erected on the buyer's foundation. Add 33% for cypress tanks. Sizes given are typical diameters and heights, in feet. Smaller tanks up to 10,000 gallons have 2" staves, larger tanks have 3". For elevated tanks, add tower cost. Add $74.00 to $84.50 per foot of diameter for sand and gravel foundation with retaining ring. Concrete slab foundation costs an additional $10.50 to $13.25 per cubic foot. Add wood cover, joists, and ladders to basic tank cost as necessary.

CAPACITY (GALLONS)	SIZE (d x h)	TANK COST	FLAT COVER	CONICAL COVER	CHIME JOISTS	WOOD LADDER	STEEL LADDER
200	4 x 3	2775.00	467.00	-----	89.00	-----	-----
300	4 x 4	3275.00	467.00	-----	89.00	-----	-----
500	5 x 4	4400.00	650.00	1300.00	130.00	-----	-----
1,000	6 x 6	6800.00	960.00	1530.00	192.00	290.00	422.00
1,500	7 x 6	8650.00	1080.00	1840.00	261.00	290.00	422.00
2,000	8 x 6	10400.00	1230.00	2120.00	323.00	290.00	422.00
3,000	8 x 8	13000.00	1230.00	2120.00	323.00	376.00	520.00
4,000	10 x 8	15600.00	1660.00	2825.00	505.00	376.00	520.00
5,000	11 x 8	18800.00	2080.00	3325.00	650.00	376.00	520.00
7,500	12 x 10	23100.00	2700.00	3800.00	745.00	467.00	615.00
10,000	14 x 10	28800.00	3125.00	4550.00	1140.00	467.00	615.00
15,000	14 x 14	41300.00	3125.00	4550.00	1140.00	635.00	830.00
20,000	16 x 14	51000.00	3825.00	6150.00	1510.00	635.00	830.00
30,000	18 x 16	66000.00	4375.00	7450.00	1910.00	695.00	940.00
50,000	22 x 18	91500.00	5500.00	9900.00	2350.00	790.00	1060.00
75,000	26 x 20	118000.00	6600.00	12300.00	2800.00	875.00	1140.00
100,000	30 x 20	143000.00	7400.00	15200.00	3650.00	875.00	1140.00
150,000	37 x 20	188000.00	10400.00	19800.00	5350.00	875.00	1140.00
200,000	43 x 20	225000.00	11900.00	24500.00	6550.00	875.00	1140.00

GALVANIZED STEEL TANKS

Costs are averages of 13 to 20 gauge, coated, corrugated steel tanks, installed or erected on the buyer's foundation. Prices include conical roof with manhole, freight and typical accessories. Deduct 15% for open tanks. Add $74.00 to $84.50 per foot of diameter for sand and gravel foundation with retaining ring. Add an additional $10.50 to $13.25 per cubic foot for concrete slab foundation. For elevated tanks, add tower cost.

CAPAC. (gallons)	DIMEN. (d x h)	TANK COST	CAPAC. (gallons)	DIMEN. (d x h)	TANK COST
500	4 x 5½	2260.00	10,000	12 x 12	22400.00
1,000	6 x 5	3800.00	15,000	15 x 11	27900.00
2,000	7 x 6½	6050.00	20,000	18½ x 11	38600.00
3,000	8 x 8	7500.00	30,000	22 x 11	53500.00
4,000	8 x 11	9100.00	50,000	24 x 15	77500.00
5,000	9 x 10	10500.00	75,000	34 x 11	114000.00
7,500	10 x 12	14000.00	100,000	34 x 15	145000.00

WOOD TOWERS

Costs are averages of painted towers for flat bottom tanks, including added cost of erection of tank above ground, footings, piping to ground, valves, balconies, ladders to balconies, and indicator gauges.

CAPAC.	TOWER HEIGHT				
(gallons)	12'	25'	50'	75'	100'
1,000	8950.00	10800.00	----	----	----
1,500	10100.00	13000.00	21200.00	----	----
2,000	11400.00	14200.00	23200.00	36800.00	----
3,000	13200.00	16500.00	26200.00	41400.00	66000.00
5,000	15400.00	19700.00	31300.00	47800.00	75500.00
10,000	20900.00	25200.00	38900.00	59000.00	89750.00
20,000	32200.00	38600.00	59250.00	88000.00	131000.00
30,000	38000.00	45300.00	67500.00	100000.00	148000.00
40,000	42100.00	50750.00	75500.00	111000.00	162000.00
50,000	----	56000.00	81250.00	118000.00	179000.00
75,000	----	----	94500.00	139000.00	217000.00

In areas subject to earthquakes or hurricanes, a rough estimate of additional cost can be obtained from the following formulas:

12' towers.....	2300.00 plus	$.06	per gallon of tank capacity
25'	3400.00 plus	$.01	per gallon of tank capacity
50'	5100.00 plus	$.14	per gallon of tank capacity
75'	8700.00 plus	$.19	per gallon of tank capacity
100'	14800.00 plus	$.25	per gallon of tank capacity

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VAULTED ABOVEGROUND PETROLEUM STORAGE TANKS

Costs are average for UL-listed cylindrical internal steel tanks encased inside a 6" precast concrete vault, providing a 4 hour fire wall and ballistic protection. The protective concrete outer shell is precast in two sections, allowing periodic internal tank inspection. Concrete support legs are cast monolithic with the lower section of the concrete vault. Costs include fittings and installation on the buyer's foundation. For supplemental internal overspill containment tank (7 to 25 gallons), add $1,930 to $4,800. For steps and platform, add $1,170 to $5,050.

CAPACITY	SIZE	SINGLE COMPARTMENT		DUAL COMPARTMENT	
(gallons)	(feet)	Single Wall	Double Wall	Single Wall	Double Wall
300	4½ x 7	22400.00	33700.00	-----	-----
550	5½ x 7	25300.00	38100.00	-----	-----
1,000	5½ x 12	32300.00	49700.00	36800.00	56000.00
2,000	7 x 14	49600.00	77000.00	54250.00	84250.00
4,000	9½ x 13	77750.00	121000.00	89750.00	140000.00
6,000	9½ x 18	99500.00	157000.00	113000.00	177000.00
8,000	9½ x 23	134000.00	209000.00	145000.00	227000.00
10,000	9½ x 29	145000.00	227000.00	157000.00	244000.00
12,000	9½ x 34	177000.00	276000.00	189000.00	296000.00

UNDERGROUND FUEL STORAGE

Costs are averages for fiberglass and steel tanks, completely installed, including fittings, access manway, excavation and backfill. For multiple installations, two or more in one hole, deduct 7% for each extra tank, considering the largest tank in the hole as the base. For difficult soil conditions, add for extra cost of excavation and bedding, as necessary.

All tanks carry Underwriters' Label. Add $10.50 to $13.25 per cubic foot for concrete pad. Costs do not include piping. Add $6,100 to $9,350 for leakage-monitoring system per tank. Sizes are approximate averages of all tank types. For multicompartment tanks, add $8,850 to $17,700 per tank.

NOMINAL CAPACITY			FIBERGLASS		STEEL (STI-P3)		FIBER COATED STEEL	
	DIAMETER	LENGTH	SINGLE WALL	DOUBLE WALL	SINGLE WALL	DOUBLE WALL	SINGLE WALL	DOUBLE WALL
(gallons)	(feet)	(feet)	TANK COST	TANK COST	TANK COST	TANK COST	TANK COST	TANK COST
300	3	5	-----	-----	12900.00	22600.00	9600.00	-----
550	4	6	12900.00	22600.00	15900.00	29100.00	12900.00	17900.00
1,000	4	11	15900.00	29100.00	20300.00	36300.00	15700.00	23500.00
2,000	6	10	20300.00	36300.00	22700.00	40400.00	19800.00	28800.00
3,000	6	13	22700.00	40400.00	25500.00	46800.00	22600.00	35700.00
4,000	7	15½	25500.00	46800.00	28800.00	50500.00	25600.00	37800.00
5,000	8	13½	28800.00	50500.00	33600.00	59000.00	28800.00	45900.00
6,000	8	18	33600.00	59000.00	37200.00	62250.00	33300.00	52250.00
8,000	8	23	37200.00	62250.00	44300.00	70000.00	37200.00	58750.00
10,000	8	29	44300.00	70000.00	49600.00	79250.00	44700.00	72250.00
12,000	8	34	49600.00	79250.00	60500.00	95750.00	51500.00	82000.00
15,000	10	29	60500.00	95750.00	79250.00	121000.00	63000.00	100000.00
20,000	10	37	79250.00	121000.00	97750.00	146000.00	82000.00	122000.00
25,000	12	33	97750.00	146000.00	117000.00	172000.00	101000.00	148000.00
30,000	12	41	117000.00	172000.00	194000.00	270000.00	120000.00	174000.00
50,000	12	60	194000.00	270000.00	172000.00	266000.00	-----	267000.00

ABOVE GROUND FUEL STORAGE

Costs are averages for U.S. listed steel tanks, completely installed, including paint, manhole if needed, skid or saddles. Sizes are approximate averages of all tank types.

CAPACITY	SINGLE WALL	DOUBLE WALL	CAPACITY	SINGLE WALL	DOUBLE WALL
(gallons)	TANK COST	TANK COST	(gallons)	TANK COST	TANK COST
550	7550.00	15700.00	8,000	39700.00	69000.00
1,000	10400.00	19800.00	10,000	43000.00	72750.00
2,000	17400.00	29400.00	12,000	45500.00	81250.00
4,000	33900.00	50250.00	15,000	56000.00	95000.00
5,000	36300.00	58500.00	20,000	60500.00	114000.00
6,000	37100.00	65000.00			

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TANKS
SECTION UIP 8

WELDED STEEL TANKS

(API)

Costs are averages for tanks erected on sand or gravel with steel ring curb, and include cone roofs with supports as needed, outside ladder, roof and shell manholes, threaded and/or flanged openings as needed for operation, roof vents, and paint. Catwalks, stairways, and platforms are not included.

CAPAC. (barrels)	SIZE (feet)	TANK COST	CAPAC. (barrels)	SIZE (feet)	TANK COST
2,000	30 x 16	241000.00	75,000	120 x 36	1892000.00
3,000	30 x 24	273000.00	100,000	140 x 37	2435000.00
4,000	30 x 32	310000.00	125,000	160 x 35	2960000.00
5,000	38 x 24	342000.00	150,000	180 x 33	3467000.00
7,500	38 x 36	395000.00	200,000	200 x 36	4231000.00
10,000	55 x 24	488000.00	250,000	220 x 36	4833000.00
15,000	55 x 36	613000.00	300,000	240 x 37	5673000.00
20,000	60 x 40	725000.00	350,000	260 x 37	6307000.00
30,000	80 x 34	954000.00	400,000	260 x 42	7042000.00
50,000	90 x 44	1352000.00	500,000	280 x 46	8394000.00

Add $2,300 to $4,200 per foot of diameter for pontoon floating roof. Add $3,025 to $3,600 per foot of diameter for double-deck roof.

HORIZONTAL BULK STORAGE

Costs are averages for completely installed steel tanks, including saddles or legs, and fittings, on owners' foundations. Typical tank dimension is diameter x length.

CAPAC. (gallons)	SIZE (feet)	TANK COST	CAPAC. (gallons)	SIZE (feet)	TANK COST
1,000	4 x 10	6400.00	7,500	7 x 26	18300.00
1,500	5 x 9	7100.00	10,000	10 x 17	22900.00
2,000	5 x 12	8100.00	12,500	10 x 21	27500.00
3,000	5 x 18	10000.00	15,000	10 x 25	32000.00
4,000	6 x 19	11700.00	20,000	10 x 34	41000.00
5,000	6 x 24	13600.00	25,000	10½ x 39	50750.00
6,000	7 x 21	15100.00	30,000	10½ x 46	58750.00

HORIZONTAL PLASTIC TANKS

Costs are averages for completely installed fiberglass or polyethylene tanks, including saddles or legs, and fittings, on owners' foundations. Sizes are approximate diameters and length sof typical tanks.

CAPAC. (gallons)	SIZE (feet)	TANK COST	CAPAC. (gallons)	SIZE (feet)	TANK COST
125	2½ x 3½	615.00	1,000	4 x 11½	2800.00
250	3 x 4	780.00	1,500	5 x 11	4425.00
500	4 x 6	1560.00	2,000	8 x 12	6200.00
750	4 x 9	2120.00	2,500	10 x 12	7650.00

VERTICAL WELDED TANKS

Costs are averages for mild steel welded tanks, including sand and gravel foundations, fittings and roof. Concrete slab foundations cost an additional $10.50 to $13.25 per cubic foot. Sizes are approximate diameters and heights of typical tanks.

CAPAC. (gallons)	SIZE (feet)	TANK COST	CAPAC. (gallons)	SIZE (feet)	TANK COST
1,500	7 x 6	10900.00	15,000	13 x 16	33700.00
2,000	7 x 8	11500.00	20,000	14 x 18	46100.00
4,000	8 x 11	15400.00	30,000	17 x 19	66000.00
5,000	9 x 11	17500.00	40,000	19 x 20	85750.00
7,500	10 x 13	22300.00	50,000	21 x 20	105000.00
10,000	11 x 15	27300.00	60,000	23 x 20	124000.00

BOLTED STEEL TANKS

(API)

Costs include roof deck and supports, sand and gravel foundation with retaining ring, painting and typical basic fittings. Sizes are approximate diameter and height of typical tanks.

CAPAC. (barrels)	SIZE (feet)	TANK COST	CAPAC. (barrels)	SIZE (feet)	TANK COST
100	9 x 8	19700.00	2,000	30 x 16	189000.00
200	9 x 16	32200.00	3,000	30 x 24	214000.00
500	16 x 16	69000.00	5,000	39 x 24	249000.00
750	16 x 24	90750.00	7,500	39 x 36	304000.00
1,000	22 x 16	113000.00	10,000	55 x 24	368000.00
1,500	22 x 24	156000.00	15,000	55 x 36	478000.00

For bulk oil loading rack platforms, see Section UIP 14.

VERTICAL BULK STORAGE

Costs are averages of 10 and 12 gauge bolted galvanized tanks, including sand and gravel foundations, fittings and roof. Concrete slab foundations cost an additional $10.50 to $13.25 per cubic foot. Typical tank dimension is diameter x height.

CAPAC. (gallons)	SIZE (feet)	TANK COST	CAPAC. (gallons)	SIZE (feet)	TANK COST
2,000	7 x 8	11100.00	15,000	13 x 16	36000.00
3,000	8 x 8	13300.00	20,000	14 x 18	44700.00
4,000	8 x 11	15200.00	30,000	17 x 19	63750.00
5,000	9 x 11	17100.00	40,000	19 x 20	82500.00
7,500	10 x 13	22000.00	50,000	21 x 20	99500.00
10,000	11 x 15	27000.00	60,000	23 x 20	120000.00

VERTICAL PLASTIC TANKS

Costs are averages for fiberglass or polyethylene tanks completely erected on buyers' foundations, including fittings and roof. Add $74 to $84.50 per foot of diameter for sand and gravel foundation with retaining ring. Concrete slab foundations cost an additional $10.50 to $13.25 per cubic foot. Sizes are approximate diameters and heights of typical tanks.

CAPAC. (gallons)	SIZE (feet)	TANK COST	CAPAC. (gallons)	SIZE (feet)	TANK COST
500	4 x 6	945.00	4,000	8 x 12	6800.00
750	4 x 9	1400.00	5,000	8½ x 12½	8450.00
1,000	5 x 7	1860.00	6,500	10 x 12	10800.00
1,500	5 x 10½	2700.00	9,000	12 x 12	15000.00
2,000	7 x 7	3200.00	12,000	12 x 14½	19700.00
2,500	8 x 7½	3875.00	16,000	14 x 14	25800.00
3,000	8 x 9	4575.00	20,000	15 x 15	31800.00

NOTE: Plastic tank prices can vary plus or minus 15% depending on the corrosive strength of the material being stored. For fiberglass underground corrosive strength chemical storage tanks see Underground Fuel Storage/Fiberglass and add 20%.

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TANKS
SECTION UIP 8

BOLTED INDUSTRIAL BINS CYLINDRICAL

The following costs are averages of bolted industrial bins for dry storage, installed without bottoms on buyers' slabs. Add $10.50 to 13.25 per cubic foot of slab.

DIAM. (feet)	HEIGHT (feet)	CAPAC. (cu. ft.)	COST to 55#/cu. ft.	COST to 80#/cu. ft.	COST to 100#/cu. ft.
9	24	1,614	15400.00	16900.00	17900.00
	32	2,152	19500.00	21100.00	22400.00
	40	2,690	22700.00	24300.00	26400.00
	56	3,776	29600.00	31000.00	34000.00
	72	4,842	35400.00	-----	-----
12	24	2,873	23800.00	25000.00	27200.00
	32	3,830	28200.00	30700.00	32200.00
	40	4,750	38500.00	40300.00	43200.00
	56	6,700	46200.00	49300.00	52250.00
	72	8,620	54000.00	-----	-----
15	24	4,485	31900.00	34200.00	36000.00
	32	5,980	39800.00	42700.00	44900.00
	48	8,970	54750.00	58250.00	61500.00
	64	11,960	69000.00	73000.00	77500.00
	80	14,966	82500.00	-----	-----
18	24	6,456	41400.00	43300.00	45600.00
	40	10,760	62000.00	65500.00	68750.00
	56	15,064	81250.00	86250.00	90500.00
	72	19,389	99250.00	104000.00	-----
	88	23,716	117000.00	-----	-----
21	32	11,725	68000.00	71500.00	75250.00
	40	14,669	80750.00	84250.00	89250.00
	56	20,531	104000.00	109000.00	114000.00
	72	26,424	125000.00	131000.00	-----
	88	32,315	147000.00	-----	-----
26	32	17,284	94500.00	98000.00	103000.00
	48	25,948	125000.00	130000.00	-----
	64	34,635	156000.00	162000.00	-----
	72	43,322	174000.00	-----	-----
	88	52,009	204000.00	-----	-----
32	32	26,378	129000	-----	-----
	40	33,006	154000	-----	-----
	56	46,264	197000	-----	-----
	72	59,521	240000	-----	-----
	88	72,778	278000	-----	-----

BOLTED INDUSTRIAL BINS HOPPERS

The following costs are averages of bolted industrial bins for dry sto age, installed with necessary structural members on concrete footings.

DIAM. (feet)	HEIGHT (feet)	CAPAC. (cu. ft.)	COST to 55#/cu. ft.	COST to 80#/cu. ft.	COST to 100#/cu. ft.
9	16	714	17200.00	19200.00	20300.00
	24	1,250	21100.00	22700.00	24000.00
	32	1,785	24000.00	25900.00	27500.00
	40	2,320	26700.00	28400.00	30100.00
12	16	1,248	25500.00	27000.00	28500.00
	24	2,200	31400.00	33700.00	36000.00
	32	3,160	35200.00	37200.00	40100.00
	40	4,115	40100.00	43300.00	46500.00
	56	6,035	47600.00	50500.00	53750.00
15	24	3,090	41200.00	43600.00	46500.00
	32	4,545	49400.00	52250.00	55750.00
	40	6,040	55750.00	59000.00	63250.00
	48	7,515	62000.00	65250.00	70250.00
	56	8,990	67250.00	71500.00	76000.00
18	24	4,235	53500.00	56500.00	60250.00
	32	6,380	64250.00	68750.00	72750.00
	40	8,535	74250.00	79000.00	84250.00
	48	10,658	82000.00	88500.00	92750.00
	56	12,850	90000.00	95750.00	102000.00
21	24	5,325	65250.00	69250.00	74250.00
	32	8,295	82000.00	87250.00	92750.00
	40	11,220	96250.00	102000.00	109000.00
	48	14,145	108000.00	114000.00	122000.00

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TANKS
SECTION UIP 8

WELDED STEEL PRESSURE TANKS

(ASME design pressure, 250 psig)

Costs are for complete standard horizontal installation on legs or saddle pads, including normal fittings on tank but not pipe, valves, or foundation. Bulk storage pressure tank prices can vary plus or minus 10% depending on shell specifications. Individual design can add 50% to 300% to the costs, depending on size. Sizes are typical nominal diameters and lengths for storage of propane, butane or ammonium sulfate, etc. For Residential use, deduct 50%

CAPACITY (gallons)	SIZE (feet)	COST	CAPACITY (gallons)	SIZE (feet)	COST
125	2 x 5½	2725.00	6,500	7 x 26	96500.00
250	2½ x 8	3600.00	9,000	7 x 35	116000.00
500	3 x 10	6500.00	12,000	7 x 45	143000.00
1,000	3½ x 15	11500.00	15,000	7 x 54	175000.00
1,500	5 x 11	17200.00	20,000	9 x 49	220000.00
2,000	5 x 15	22600.00	30,000	11 x 47	307000.00
2,500	5 x 19	27900.00	45,000	11 x 63	438000.00
3,000	5 x 22	30500.00	60,000	11 x 90	568000.00
4,000	5 x 29	40000.00	90,000	11 x 133	835000.00

FABRIC PILLOW TANKS

Average costs of rubber or vinyl coated fabric pillow tanks, which lay flat when empty and assume a “pillow” configuration when full of liquid. Costs include typical tank ancillaries. Sitework and exterior piping are extra.

CAPACITY (gallons)	SIZE (feet)	POTABLE WATER	FUELS, OILS, CHEMICALS AND MILD SOLUTIONS
100	6 x 4	1270.00	2370.00
250	9 x 5	1650.00	2875.00
500	10 x 7	2120.00	3525.00
1,000	12 x 9	2875.00	4650.00
2,000	13 x 12	4275.00	6800.00
3,000	14 x 14	5200.00	8250.00
5,000	17 x 16	7100.00	11000.00
7,500	20 x 19	8900.00	13800.00
10,000	21 x 22	10500.00	16400.00
15,000	24 x 25	13100.00	19900.00
20,000	30 x 25	16100.00	23900.00
25,000	36 x 25	18700.00	27600.00
50,000	65 x 25	34900.00	51750.00
100,000	85 x 35	58250.00	90250.00

LARGE PRESSURE TANKS

Costs are averages including erection, structural supports, normal foundations and appurtenant equipment. These tanks are all individ ally designed, and costs vary greatly. Sphere wall thickness will cause costs to vary plus or minus 20%. For areas requiring earthquake- or hurricane-resistant structures, increase costs by 15%.

SPHERES

DIAMETER (feet)	CAPAC. (cu. ft.)	COST	DIAMETER (feet)	CAPAC. (cu. ft.)	COST
20	4,190	281000.00	40	33,510	826000.00
25	8,180	397000.00	45	47,715	990000.00
30	14,135	529000.00	50	65,450	1161000.00
35	22,450	671000.00	60	113,095	1546000.00

HEMISPHEROIDS

CAPACITY	5 LB. W.P.	10 LB. W.P.	25 LB. W.P.
105,000 gallons	391000.00	452000.00	525000.00
210,000 gallons	557000.00	658000.00	793000.00
420,000 gallons	806000.00	963000.00	1198000.00
840,000 gallons	1149000.00	1401000.00	1814000.00

DEWAR TANKS (CYROGENIC, 250 LB. W.P.) VERTICAL

CAPACITY	COST	CAPACITY	COST
500 gallons	79250.00	6,000 gallons	215000.00
1,000 gallons	106000.00	9,000 gallons	296000.00
3,000 gallons	147000.00	12,000 gallons	349000.00

HORIZONTAL

CAPACITY	COST	CAPACITY	COST
500 gallons	87000.00	6,000 gallons	272000.00
1,000 gallons	119000.00	9,000 gallons	368000.00
3,000 gallons	169000.00	12,000 gallons	438000.00

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MISCELLANEOUS INDUSTRIAL COSTS

SECTION UIP 9

INDUSTRIAL BOILERS

Costs are for gas and light oil fired boilers, complete with pumps, controls and gauges, 15 pounds pressure for steam, 30 pounds for water. Costs do not include flue piping, electric wiring, pad, gas or oil piping.

Add 12% for 125# water pressure, 25% for 150# steam.

For heavy-oil- and combination oil- and gas fired units, add 15%.

HP	WATER TUBE	SCOTCH MARINE	FIRE TUBE
4	6100.00	-----	-----
6	8550.00	-----	16100.00
10	12400.00	-----	21700.00
15	16100.00	22500.00	27900.00
20	19700.00	26400.00	32600.00
30	26700.00	34100.00	41900.00
40	31400.00	39600.00	48600.00
50	37100.00	44800.00	56000.00
75	49700.00	57500.00	70250.00
100	61750.00	68000.00	83500.00
150	83500.00	87500.00	106000.00
200	100000.00	104000.00	127000.00
300	136000.00	130000.00	159000.00
400	167000.00	155000.00	189000.00
500	203000.00	179000.00	217000.00
600	226000.00	201000.00	244000.00

Large high capacity packages boilers will cost $525 to $710 per horsepower.

Examples: 1,000 HP x $710 = $710,000;
6,000 HP x $525 = $3,150,000.

DEFINITIONS

WATER TUBE is a boiler in which the tubes contain water and steam, with the heat being applied to the outside surface. These are generally small commercial boilers with steel or cast iron bodies.

SCOTCH MARINE is a cylindrical steel shell “fire tube” boiler with one or more cylindrical internal steel furnaces located generally in the lower portion, with a bank or banks (passes) of tubes attached to both end closures.

FIRE TUBE refers to a boiler with straight tubes, which are surrounded by water and steam and through which the products of combustion pass.

BOILER RATINGS

Packaged boilers are rated on the basis of maximum continuous nozzle output, in terms of boiler horsepower, B.T.U. per hour, or pounds of steam per hour at 212° F. Available information can be converted to horsepower by using the table below.

1 boiler horsepower = 33,500 B.T.U. per hour
= 139 square feet of steam radiation
= 223 square feet of water radiation
= 34.5# of steam per hour

1 pound of steam per hour = 970 B.T.U. per hour
1 square foot of equivalent steam radiation = 240 B.T.U. per hour
1 square foot of equivalent water radiation = 150 B.T.U. per hour

INDUSTRIAL PUMPS

The costs listed here are only indications of the average installed costs, since there are too many types of pumps and service conditions to provide really accurate estimates. This list contains a few basic centrifugal pumps for water or the equivalent, in normal temperature ranges. The cost includes drip-proof motor, impeller, coupling and case.

SUCTION	DISCHARGE	HP	3550 RPM	1750 RPM	1160 RPM
1¼”	x 1”	1½	2575.00	-----	-----
1½”	x 1¼”	1½	3225.00	3225.00	-----
2”	x 1½”	3	3700.00	4125.00	-----
		5	4275.00	4550.00	-----
		7½	5300.00	5400.00	-----
2½”	x 2”	5	3300.00	5000.00	-----
		7½	5350.00	5750.00	-----
3”	x 2½”	5	4700.00	4850.00	-----
		7½	5550.00	5750.00	-----
		10	5900.00	6250.00	-----
4”	x 2”	5	4775.00	5150.00	-----
		10	6100.00	6850.00	-----
		15	7350.00	8650.00	-----
		20	7900.00	9600.00	-----
4”	x 3”	5	5200.00	5900.00	-----
		10	6550.00	7300.00	-----
		15	7800.00	8850.00	-----
5”	x 4”	10	-----	7750.00	-----
		15	-----	8850.00	-----
		20	-----	9700.00	-----
		30	10400.00	11900.00	-----
6”	x 4”	40	-----	13900.00	-----
		50	-----	15600.00	-----
		60	13100.00	18300.00	22000.00
		75	17400.00	21100.00	25000.00
6”	x 5”	20	-----	10500.00	-----
		30	-----	14400.00	12200.00
		50	-----	16500.00	19700.00
		75		21100.00	-----
8”	x 6”	40	-----	15200.00	-----
		60	-----	19500.00	21600.00
		75	-----	22900.00	-----
		100	-----	26400.00	-----
		125	-----	28300.00	-----

For costs of pumps with horsepower ratings not listed, use the proper suction and discharge size and adjust for the cost of the proper motor from Page 6, or interpolate. Add 50% for horizontal split-case pumps.

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PIPING
SECTION UIP 9

GENERAL INFORMATION

The pipe costs on this and the following page are averages of installed costs per linear foot including contractors' overhead and profit, but e cluding any design layout costs or fees. All sizes refer to interior diameter of the pipe.

The costs are listed under two broad categories: service piping, for pipe, fittings and valves within the building lines, and utility piping, for pipes, fittings and valves installed outside and up to the building lines.

Each category is further subdivided into pressure lines and nonpre sure lines. Pressure lines generally carry water, gas, steam, etc., under constant pressure, while non-pressure lines, used for drain, waste and venting, are not subject to pressure from the materials they carry.

SERVICE PIPING

Costs for pipe under service piping represent the plumbing, HVAC and processing systems, ascending in that order. Typical fittings are included, but valves, hangers and supports, and trenching and backfill, which are all listed separately, must be added if needed.

For small or intricate installations, costs may be 15% to 25% higher. For long, straight runs with minimum fittings, costs may be 15% to 25% lower. For piping with mixed materials (e.g. steel with plastic lining) use higher end of listed costs.

For galvanized steel pipe, add 5% to 10% to black steel pipe costs. Use welded joint pipe costs for victaulic coupled or flanged steel pipe.

SERVICE PIPE

PRESSURE PIPE	¼"	½"	¾"	1"
Copper	11.00 – 12.60	12.85 – 15.50	15.10 – 18.70	18.40 – 23.15
Glass	----- – -----	19.80 – 37.50	25.00 – 42.75	31.75 – 50.00
Plastic	9.16 – 9.49	9.59 – 10.60	10.00 – 11.00	11.00 – 11.80
Black steel (threaded)	----- – -----	12.6 – 14.15	15.1 – 18.00	18.30 – 22.70
Steel (welded joint)	----- – -----	----- – -----	----- – -----	26.25 – 32.25
Add for trench and fill	3.28 – 3.77	3.54 – 4.36	3.77 – 4.71	3.86 – 5.47

PRESSURE PIPE	1¼"	1½"	2"	2½"
Copper	21.00 – 31.75	26.00 – 35.50	37.00 – 45.00	55.00 – 67.00
Glass	----- – -----	16.90 – 63.50	56.50 – 76.50	----- – -----
Plastic	12.70 – 13.80	13.50 – 15.05	15.65 – 17.15	17.35 – 19.65
Black steel (threaded)	21.00 – 29.75	24.60 – 34.75	30.25 – 45.00	36.00 – 56.00
Steel (welded joint)	32.25 – 39.75	33.25 – 42.50	40.25 – 51.50	52.00 – 66.00
Add for trench and fill	4.29 – 5.97	4.36 – 7.91	4.78 – 7.57	5.47 – 8.16

PRESSURE PIPE	3"	4"	6"	8"
Copper	63.50 – 79.00	108.00 – 138.00	177.00 – 232.00	320.00 – 432.00
Glass	73.50 – 103.00	93.00 – 141.00	109.00 – 225.00	----- – -----
Plastic	20.85 – 24.00	26.75 – 31.50	39.50 – 51.50	48.75 – 61.00
Black steel (threaded)	42.75 – 67.50	57.00 – 93.00	118.00 – 159.00	200.00 – 240.00
Steel (welded joint)	61.50 – 79.00	96.00 – 125.00	125.00 – 170.00	148.00 – 200.00
Add for trench and fill	5.97 – 9.00	6.99 – 10.75	8.59 – 14.65	10.60 – 18.40

PRESSURE PIPE	10"	12"	16"	24"
Copper	----- – -----	----- – -----	----- – -----	----- – -----
Glass	----- – -----	----- – -----	----- – -----	----- – -----
Plastic	64.00 – 82.00	82.00 – 108.00	----- – -----	----- – -----
Black steel (threaded)	281.00 – 335.00	368.00 – 416.00	----- – -----	----- – -----
Steel (welded joint)	185.00 – 249.00	217.00 – 289.00	289.00 – 393.00	560.00 – 790.00
Add for trench and fill	12.55 – 21.95	14.40 – 25.50	18.25 – 33.25	25.75 – 47.75

DRAIN, WASTE AND VENT	1¼"	1½"	2"	4"
Cast iron	----- – -----	17.45 – 28.00	26.25 – 39.75	39.50 – 54.00
Copper	21.15 – 29.75	22.80 – 32.00	27.50 – 39.00	61.00 – 88.50
Glass	----- – -----	32.75 – 54.50	42.75 – 62.00	93.00 – 106.00
Plastic (standard)	11.95 – 13.40	13.40 – 14.85	14.85 – 16.60	24.70 – 29.00
Plastic (acid waste)	----- – -----	18.15 – 33.25	25.50 – 39.00	55.00 – 71.50

DRAIN, WASTE AND VENT	6"	8"	10"	12"
Cast iron	56.00 – 72.00	90.00 – 122.00	133.00 – 159.00	185.00 – 210.00
Copper	144.00 – 217.00	383.00 – 595.00	----- – -----	----- – -----
Glass	159.00 – 185.00	----- – -----	----- – -----	----- – -----
Plastic (standard)	32.25 – 45.00	----- – -----	----- – -----	----- – -----
Plastic (acid waste)	62.00 – 90.00	----- – -----	----- – -----	----- – -----

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PIPING
SECTION UIP 9

SERVICE PIPE VALVES

Valve costs are averages of many types and are listed by material composition and size of pipe they serve. The costs are divided into three groups: general service (under 150 PSI), medium duty (150 - 300 PSI) and heavy duty service (over 300 PSI). These three classif cations are used as descriptions of the pressure ratings of the valves and not as technical specifications. Most plumbing and H.V.A.C. valves will be priced under general service. Medium and heavy duty valves generally represent steam and other industrial system valves only.

VALVES, EACH	¼"	½"	¾"	1"
Bronze, general	62.50 – 94.50	75.00 – 111.00	101.00 – 136.00	114.00 – 170.00
medium	76.50 – 103.00	82.00 – 133.00	109.00 – 170.00	124.00 – 258.00
Iron body, general	----- – -----	----- – -----	----- – -----	----- – -----
medium	----- – -----	----- – -----	----- – -----	----- – -----
Plastic, general	73.50 – 130.00	82.00 – 152.00	103.00 – 185.00	114.00 – 225.00
Steel, general (threaded)	----- – -----	----- – -----	185.00 – 258.00	281.00 – 401.00
medium	----- – -----	----- – -----	----- – -----	----- – -----

VALVES, EACH	1¼"	1½"	2"	2½"
Bronze, general	133.00 – 210.00	159.00 – 258.00	232.00 – 289.00	328.00 – 615.00
medium	158.00 – 352.00	185.00 – 456.00	265.00 – 640.00	416.00 – 940.00
Iron body, general	----- – -----	----- – -----	665.00 – 915.00	760.00 – 1010.00
medium	----- – -----	----- – -----	870.00 – 1130.00	1150.00 – 1580.00
Plastic, general	136.00 – 265.00	159.00 – 311.00	225.00 – 480.00	441.00 – 915.00
Steel, general (threaded)	360.00 – 496.00	408.00 – 575.00	489.00 – 745.00	----- – -----
medium	----- – -----	----- – -----	1530.00 – 2825.00	2130.00 – 3175.00

VALVES, EACH	3"	4"	6"	8"
Iron body, general	855.00 – 1120.00	1080.00 – 1370.00	1730.00 – 2170.00	2675.00 – 3550.00
medium	1420.00 – 1940.00	2030.00 – 2900.00	3600.00 – 5450.00	5300.00 – 8050.00
Steel, general (flanged)	1730.00 – 3100.00	2320.00 – 3950.00	3950.00 – 6650.00	6800.00 – 11000.00
medium	2675.00 – 3625.00	3900.00 – 4650.00	6800.00 – 7650.00	10000.00 – 12800.00
heavy duty	4875.00 – 5850.00	10600.00 – 12300.00	13900.00 – 15900.00	21400.00 – 24700.00

VALVES, EACH	10"	12"	16"	24"
Iron body, general	4425.00 – 5600.00	7050.00 – 8850.00	----- – -----	----- – -----
medium	7200.00 – 11300.00	9150.00 – 14600.00	----- – -----	----- – -----
Steel, general (flanged)	9450.00 – 16700.00	12200.00 – 22800.00	20700.00 – 38300.00	57250.00 – 78500.00
medium	13700.00 – 19200.00	17300.00 – 27100.00	43600.00 – 47400.00	----- – -----
heavy duty	29800.00 – 34100.00	38700.00 – 45000.00	----- – -----	----- – -----

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PIPING
SECTION UIP 9

SERVICE PIPE HANGERS AND SUPPORTS

HANGERS, per linear foot	¼"	½"	¾"	1"
Cast iron	----- – -----	----- – -----	----- – -----	----- – -----
Copper	3.10 – 3.69	3.96 – 4.78	4.71 – 5.97	5.56 – 6.81
Glass	----- – -----	5.90 – 7.13	6.05 – 7.33	6.24 – 7.40
Plastic	4.78 – 6.05	5.29 – 6.64	5.90 – 6.99	6.24 – 7.91
Steel (threaded)	2.70 – 3.10	3.10 – 3.35	3.35 – 3.96	3.96 – 4.36
Steel (welded joint)	----- – -----	----- – -----	----- – -----	5.47 – 6.24

HANGERS, per linear foot	1¼"	1½"	2"	2½"
Cast iron	----- – -----	5.90 – 6.46	6.05 – 6.81	6.57 – 7.40
Copper	6.12 – 7.74	6.57 – 8.16	7.33 – 9.16	8.25 – 10.25
Glass	----- – -----	6.64 – 7.82	7.22 – 8.33	----- – -----
Plastic	6.64 – 8.25	6.99 – 9.07	8.09 – 9.93	8.82 – 11.35
Steel (threaded)	4.36 – 4.78	4.71 – 5.38	5.56 – 6.12	6.64 – 7.22
Steel (welded joint)	5.97 – 6.73	6.24 – 7.22	6.99 – 8.16	7.91 – 9.07

HANGERS, per linear foot	3"	4"	6"	8"
Cast iron	6.99 – 7.74	7.74 – 8.59	9.49 – 10.45	11.70 – 12.70
Copper	9.07 – 11.35	10.35 – 13.05	12.45 – 15.95	----- – -----
Glass	8.09 – 9.16	8.91 – 10.25	10.70 – 13.40	----- – -----
Plastic	9.25 – 11.95	11.45 – 14.55	14.90 – 18.75	18.55 – 23.45
Steel (threaded)	6.99 – 7.74	8.59 – 9.49	11.85 – 13.05	15.05 – 16.55
Steel (welded joint)	8.59 – 9.93	10.25 – 11.85	13.50 – 15.95	16.60 – 19.80

HANGERS, per linear foot	10"	12"	16"	24"
Cast iron	14.20 – 15.70	17.20 – 19.35	----- – -----	----- – -----
Copper	----- – -----	----- – -----	----- – -----	----- – -----
Glass	----- – -----	----- – -----	----- – -----	----- – -----
Plastic	21.75 – 27.75	25.50 – 32.25	----- – -----	----- – -----
Steel (threaded)	18.55 – 20.10	21.60 – 23.60	----- – -----	----- – -----
Steel (welded joint)	20.05 – 23.40	23.15 – 27.75	29.75 – 35.25	43.25 – 50.50

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PIPING

SECTION UIP 9

UTILITY PIPING

Cost per linear foot for underground utility lines, including fittings, an allowance for trenching and backfill and contractors' overhead and profit. For non circular pipe, use the average diameter of the smallest and largest dimension.

PRESSURE PIPE	4"	6"	8"	10"
Asbestos cement	44.75 – 56.50	55.50 – 72.00	71.50 – 91.50	118.00 – 146.00
Ductile iron	48.00 – 59.00	53.00 – 67.00	82.00 – 105.00	99.00 – 130.00
Concrete	----- – -----	----- – -----	----- – -----	----- – -----
Plastic	25.75 – 33.75	30.50 – 39.50	45.50 – 58.00	55.50 – 79.00
Steel	56.50 – 62.00	69.00 – 78.00	87.00 – 103.00	129.00 – 155.00
Valves, each	1170.00 – 1520.00	2575.00 – 3125.00	4500.00 – 5300.00	7050.00 – 8200.00

PRESSURE PIPE	12"	14"	16"	18"
Asbestos cement	136.00 – 170.00	147.00 – 210.00	192.00 – 249.00	240.00 – 335.00
Ductile iron	111.00 – 147.00	129.00 – 177.00	158.00 – 217.00	192.00 – 265.00
Concrete	----- – -----	----- – -----	73.00 – 93.00	88.50 – 109.00
Plastic	82.00 – 103.00	----- – -----	----- – -----	----- – -----
Steel	148.00 – 185.00	156.00 – 192.00	185.00 – 232.00	225.00 – 289.00
Valves, each	9450.00 – 11500.00	12400.00 – 15200.00	16100.00 – 19500.00	20100.00 – 24600.00

PRESSURE PIPE	20"	24"	30"	36"
Asbestos cement	273.00 – 383.00	311.00 – 456.00	393.00 – 575.00	525.00 – 790.00
Cast iron	210.00 – 289.00	225.00 – 311.00	258.00 – 360.00	320.00 – 463.00
Concrete	93.00 – 122.00	124.00 – 159.00	177.00 – 240.00	232.00 – 320.00
Steel (treated & wrapped)	240.00 – 311.00	281.00 – 368.00	352.00 – 480.00	489.00 – 675.00
Valves, each	24600.00 – 29500.00	34700.00 – 43200.00	53250.00 – 63500.00	74500.00 – 89750.00

PRESSURE PIPE	42"	48"	60"	72"
Asbestos cement	----- – -----	----- – -----	----- – -----	----- – -----
Cast iron	383.00 – 565.00	441.00 – 650.00	----- – -----	----- – -----
Concrete	289.00 – 401.00	352.00 – 505.00	505.00 – 720.00	665.00 – 975.00
Steel (treated & wrapped)	675.00 – 945.00	750.00 – 1070.00	----- – -----	----- – -----
Valves, each	99750.00 – 121000.00	130000.00 – 153000.00	----- – -----	----- – -----

DRAIN AND SEWER	4"	6"	8"	10"
Asbestos cement	----- – -----	22.35 – 29.00	26.00 – 33.00	48.75 – 62.50
Corrugated metal	----- – -----	26.25 – 30.25	34.25 – 41.75	52.00 – 65.50
Plastic	10.75 – 11.85	13.85 – 18.40	21.00 – 28.50	27.75 – 39.75
Plain concrete	----- – -----	27.00 – 29.00	33.75 – 38.25	41.00 – 48.50
Reinforced concrete	----- – -----	----- – -----	----- – -----	----- – -----
Vitrified clay	19.80 – 27.75	29.75 – 36.75	39.50 – 48.00	70.50 – 79.50

DRAIN AND SEWER	12"	15"	18"	21"
Asbestos cement	54.00 – 68.00	69.50 – 90.00	101.00 – 129.00	111.00 – 141.00
Corrugated metal	61.00 – 69.50	66.50 – 83.50	73.50 – 99.00	94.50 – 127.00
Plastic	34.00 – 53.50	45.00 – 82.00	----- – -----	----- – -----
Plain concrete	46.50 – 56.50	49.50 – 62.00	58.50 – 75.00	73.50 – 96.00
Reinforced concrete	48.00 – 62.00	53.00 – 68.50	67.50 – 88.50	90.00 – 117.00
Vitrified clay	78.00 – 91.50	106.00 – 124.00	138.00 – 159.00	170.00 – 200.00

DRAIN AND SEWER	24"	30"	36"	42"
Asbestos cement	122.00 – 156.00	146.00 – 185.00	210.00 – 265.00	----- – -----
Corrugated metal	105.00 – 141.00	120.00 – 153.00	185.00 – 258.00	217.00 – 295.00
Reinforced concrete	109.00 – 144.00	124.00 – 159.00	192.00 – 265.00	249.00 – 328.00
Vitrified clay	210.00 – 249.00	249.00 – 295.00	328.00 – 408.00	----- – -----

DRAIN AND SEWER	48"	54"	60"	72"
Asbestos cement	----- – -----	----- – -----	----- – -----	----- – -----
Corrugated metal	265.00 – 352.00	335.00 – 463.00	432.00 – 600.00	480.00 – 665.00
Reinforced concrete	289.00 – 360.00	343.00 – 480.00	448.00 – 615.00	545.00 – 735.00
Vitrified clay	----- – -----	----- – -----	----- – -----	----- – -----

DRAIN AND SEWER	78"	84"	90"	96"
Corrugated metal	525.00 – 720.00	565.00 – 765.00	610.00 – 825.00	655.00 – 885.00
Reinforced concrete	655.00 – 870.00	750.00 – 985.00	820.00 – 1090.00	910.00 – 1190.00

DRAIN AND SEWER	102"	108"	114"	120"
Corrugated metal	705.00 – 945.00	745.00 – 990.00	790.00 – 1060.00	855.00 – 1110.00
Reinforced concrete	----- – -----	----- – -----	----- – -----	----- – -----

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INDUSTRIAL MOTORS
SECTION UIP 9

EXPLANATION

The following table includes average costs of completely installed standard motors, including connections and anchors, based on normal industrial installations of 500 to 1,500 total horsepower. For variations and adjustments, see notes and example below table. High efficiency motors can run 20% to 50% more.

POWER WIRING

For total electrical cost estimates, the average cost of power wiring for a 480-volt system is listed. Costs include wiring, starters and/or control panels. The cost for 230-volt power wiring will be up to 75% more than a 480-volt system in normal industrial installations.

ELECTRIC MOTORS

	1800-RPM MOTORS					4:1 VARIABLE-DRIVE MOTORS			
RATED HP	DRIP-PROOF	TOTALLY ENCLOSED	EXPLOSION-PROOF	480-VOLT POWER WIRING	ADD FOR SLIDE RAILS	DRIP-PROOF	TOTALLY ENCLOSED	EXPLOSION-PROOF	RATED HP
½	625.00	720.00	1030.00	790.00	118.00	3025.00	3100.00	3300.00	½
1	835.00	910.00	1140.00	855.00	118.00	3100.00	3275.00	3575.00	1
1½	955.00	1030.00	1190.00	855.00	118.00	3225.00	3300.00	3625.00	1½
2	1050.00	1140.00	1280.00	930.00	127.00	3300.00	3575.00	3700.00	2
3	1170.00	1350.00	1390.00	1060.00	143.00	3650.00	3700.00	4100.00	3
5	1390.00	1580.00	1730.00	1390.00	192.00	4125.00	4250.00	4550.00	5
7½	1580.00	1790.00	2110.00	1650.00	200.00	4700.00	4875.00	5300.00	7½
10	1790.00	1970.00	2490.00	1970.00	200.00	5300.00	5850.00	6150.00	10
15	2230.00	2650.00	3350.00	2525.00	210.00	6250.00	6900.00	7350.00	15
20	2650.00	3475.00	4275.00	3225.00	281.00	7350.00	7950.00	8650.00	20
25	3100.00	4250.00	5250.00	3900.00	311.00	8650.00	9150.00	9700.00	25
30	3650.00	4875.00	6150.00	4550.00	352.00	9700.00	10100.00	10900.00	30
40	4575.00	6500.00	7950.00	5750.00	368.00	12000.00	12200.00	13100.00	40
50	5550.00	7950.00	9850.00	8700.00	393.00	14400.00	14400.00	15300.00	50
75	8650.00	13300.00	18300.00	10300.00	408.00	20000.00	-----	-----	75
100	11600.00	18300.00	25300.00	13400.00	463.00	26100.00	-----	-----	100
125	14600.00	23200.00	31700.00	16400.00	555.00	-----	-----	-----	125
150	17900.00	27900.00	36200.00	20000.00	555.00	-----	-----	-----	150
200	24400.00	38000.00	43600.00	26100.00	615.00	-----	-----	-----	200
250	31000.00	47500.00	49400.00	32300.00	650.00	-----	-----	-----	250

ADJUSTMENTS

CONSTANT-SPEED MOTORS

For constant motor speeds other than 1800 RPM follow the following rules:

1200 RPM	Use the price of the next larger size motor.
900 RPM	Use the price of a motor two sizes larger.
720 RPM	Use the price of the third size larger.
3600 RPM	Use the price of the next smaller size motor for motors between 5 and 75 HP

Under 5 HP and over 100 HP, use the price of the next larger motor.

VARIABLE-DRIVE MOTORS

Variable drive motors come in a great many combinations of output speeds, and the actual costs will vary from those above by a minus 30 percent for extremely high-speed (780-RPM) output to a plus 60 percent for an extremely low-speed (45-RPM) output. The costs above are averages for typical motors with maximum outputs of approximately 100 to 400 RPM. For more accurate figures, it is recommended that manufacturers' costs be obtained.

EXAMPLE

5 HP	1800 RPM, dripproof	1390.00
5 HP	1200 RPM, dripproof	1580.00
5 HP	900 RPM, dripproof	1790.00
5 HP	720 RPM, dripproof	2230.00
5 HP	3600 RPM, dripproof	1170.00

COST MULTIPLIERS – VARIABLE DRIVE

For drive ratios other than 4:1, use the following cost multipliers:

Ratio 2:1, multiply by 1.07 Ratio 3:1, multiply by 1.03

Ratio 5:1, multiply by .98

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MISCELLANEOUS INDUSTRIAL STRUCTURES

SECTION UIP 9

STEEL STACKS

(Average costs per foot of height, including foundation; painted outside only)

THICKNESS (AT BASE)	18"	24"	30"	36"	42"	48"	THICKNESS (at base)
12 gauge	265.00	335.00	408.00	480.00	545.00	615.00	12 gauge
10 gauge	311.00	393.00	480.00	560.00	635.00	710.00	10 gauge
8 gauge	360.00	456.00	545.00	625.00	730.00	820.00	8 gauge
3/16"	393.00	496.00	595.00	705.00	780.00	880.00	3/16"
1/4"	-----	595.00	705.00	805.00	910.00	1010.00	1/4"
5/16"	-----	-----	805.00	925.00	1060.00	1150.00	5/16"
3/8"	-----	-----	-----	1010.00	1150.00	1280.00	3/8"
1/2"	-----	-----	-----	-----	1340.00	1470.00	1/2"
Guy wires (ln. ft.)*	0.85	0.97	1.07	1.22	1.36	1.39	Guy wires (ln. ft)*
Guy bands (each)	90.00	135.00	170.00	210.00	258.00	289.00	Guy bands (each)
Roof flashing	610.00	885.00	1140.00	1410.00	1710.00	2030.00	Roof flashing
Umbrella top (each)	232.00	343.00	480.00	625.00	780.00	955.00	Umbrella top (each)

THICKNESS (at base)	54"	60"	66"	72"	78"	84"	THICKNESS (at base)
12 gauge	705.00	765.00	835.00	-----	-----	-----	12 gauge
10 gauge	790.00	880.00	945.00	1000.00	-----	-----	10 gauge
8 gauge	895.00	970.00	1060.00	1130.00	1190.00	-----	8 gauge
3/16"	955.00	1060.00	1130.00	1230.00	1310.00	1340.00	3/16"
1/4"	1110.00	1230.00	1330.00	1410.00	1470.00	1590.00	1/4"
5/16"	1270.00	1360.00	1470.00	1590.00	1700.00	1760.00	5/16"
3/8"	1410.00	1530.00	1590.00	1760.00	1880.00	1940.00	3/8"
1/2"	1590.00	1760.00	1880.00	2010.00	2160.00	2280.00	1/2"
Guy wires (ln. ft.)*	1.46	1.64	1.71	1.77	1.82	1.91	Guy wires (ln. ft)*
Guy bands (each)	335.00	383.00	432.00	463.00	510.00	565.00	Guy bands (each)
Roof flashing	2280.00	2625.00	2925.00	3225.00	3550.00	3875.00	Roof flashing
Umbrella top (each)	-----	-----	-----	-----	-----	-----	Umbrella top (each)

For double wall band type, prefabricated refractory lined, add 25% to 50%; welded jacket type, add 75% to 100%.
*For guy cable, add cost per strand. For stainless steel cable, add 100%.

BRICK AND CONCRETE STACKS

Average costs per foot of height, including foundations up to 100 feet in height. Add 1% for each additional foot of height to all costs up to 200'; over 200' add .25% per foot. For square or rectangular chimneys, use 1/3 the perimeter in place of the diameter. For refractory lining, increase cost by 100% to 200% of lined area.

BASE DIAM. (OUTSIDE)	BRICK	CON-CRETE	BASE DIAM. (OUTSIDE)	BRICK	CON-CRETE
6'	1360.00	1160.00	16'	3025.00	2600.00
7'	1560.00	1280.00	18'	3350.00	2925.00
8'	1730.00	1430.00	20'	3600.00	3200.00
9'	1950.00	1630.00	22'	3925.00	3525.00
10'	2100.00	1730.00	24'	4250.00	3775.00
12'	2480.00	2060.00	28'	4800.00	4250.00
14'	2775.00	2410.00	32'	5500.00	4875.00

CHIMNEYS

MASONRY: Average cost per foot of height with tile flues, including foundation. For custom designs, add 50% to 100%.

8" square or round flues	130.00 – 220.00
12" square or round flues	142.00 – 238.00
2 – 8" square or round flues	205.00 – 350.00
10" x 18" rectangular flue	197.00 – 331.00
1 – 8" and 1 – 12" square flues	229.00 – 396.00

PREFABRICATED METAL: Average cost per linear foot for round galvanized metal stacks including ancillary items. Add $320 to $560 for thru the wall installations. For box framed wood or stucco decorative chimney housing add $33.75 to $50.50 per linear foot and $67.50 to $100.00 for metal units.

SIZE	SINGLE WALL	DOUBLE WALL	TRIPLE WALL
6"	38.75 – 57.50	45.00 – 67.00	59.00 – 87.00
8"	41.75 – 63.50	50.50 – 75.00	73.50 – 111.00
10"	45.50 – 67.50	64.00 – 99.00	94.50 – 135.00

FIREPLACES

	COST RANGE
Metal, prefab. fireboxes or forms up to 48"	1180.00 - 2380.00
see through, 3 sided or extra large openings	2650.00 - 4775.00
Masonry, brick firebox	4000.00 - 5300.00
Hearth	885.00 - 1200.00
Custom, extra large opening	6650.00 - 16000.00
Hearth	1150.00 - 1990.00
Soapstone heaters, complete, including chimney	13300.00 - 53250.00
Add for glass doors	437.00 - 655.00
Circulation blowers, each	229.00 - 372.00
Draft inducer fans	3125.00 - 4725.00

MANTELS

Prefabricated, wood or fiberglass	372.00 - 1540.00
Custom, special design or antique reproductions	
Cast stone (bonded limestone)	6250.00 - 18700.00
Marble, granite or onyx	11500.00 - 42700.00
Ornate wood	6250.00 - 8850.00
Precast plaster or flax decorated	3200.00 - 7850.00

For masonry veneer wall facing, see Section UIP 5.

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MISCELLANEOUS INDUSTRIAL COSTS

SECTION UIP 9

METAL WASTE CHUTES

Average cost per foot of height, including doors, roof vent and sprinklers. For electrically operated doors, add $1,730 to $2,280 each. For stainless steel, add 25%.

Diameter (inside)	15”	18”	21”	24”	30”	36”
Low Cost	150.00	170.00	177.00	192.00	217.00	240.00
Average Cost	177.00	200.00	217.00	240.00	273.00	304.00
High Cost	217.00	240.00	265.00	289.00	335.00	375.00

COMPACTORS
COMMERCIAL OR INDUSTRIAL – STATIONARY

CAPAC. (CU. YD.)	COST	CAPAC. (CU. YD.)	COST
1/2 and under	10500.00 - 17900.00	3	46300.00 - 55000.00
1	22000.00 - 27400.00	5	77500.00 - 86750.00
2	34400.00 - 41100.00	7	114000.00 - 127000.00

NOTE: For building chute-fed compactors, add $2,310 to $9,250 for each container; small bag chute compactors, add 50% to costs above.

DRUM COMPACTORS

Costs are average for compactors with cylindrical chambers, used to crush pails or drums and to compact within drum waste material. The costs include totally enclosed, fan cooled motors and starters, NEMA 1 control cabinets and 115-volt, pushbutton controls. Costs include installation. For compactors with pneumatic controls, cost of 60-psi air is not included.

COMPACTION FORCE (POUNDS)	COMPACTION (HEIGHT)	CHAMBER (DIAMETER)	ELECTRIC MOTOR (HP)	COST
12,000	17”	16”	3	17300.00
20,000	48”	38”	7.5	43500.00
50,000	50”	32”	10	48100.00
60,000	38”	32”	10	58000.00
85,000	38”	32”	10	70000.00
85,000	50”	36”	10	79250.00

For explosion proof motors and pneumatic or hydraulic controls, add 20% to 30%.

For drum roll-out dolly and platform add $3,675.

Air filter systems connected to compactors above, to filter airborne matterfrom the compaction chambers cost $22,300. Air filter systems meeting ANSI-N510 hazardous air filtration standards cost $82,750.

INCINERATORS
COMMERCIAL OR INDUSTRIAL – STEEL

Costs do not include scrubber, chimney, electrical panel or piping. Add $19,400 to $29,500 for feeder.

Pound per Hour	Cost	Pound per Hour	Cost
50	38600.00 – 48300.00	400	167000.00 – 214000.00
100	61250.00 – 69250.00	600	222000.00 – 298000.00
200	94250.00 – 104000.00	1,000	259000.00 – 349000.00

MASONRY

Costs include brickwork but do not include chimney or air pollution control. For refractory lining add 200%.

Pound per Hour	Cost	Pound per Hour	Cost
100	16100.00 – 21100.00	1,000	55500.00 – 61000.00
200	20300.00 – 25800.00	2,000	99000.00 – 111000.00
400	29800.00 – 33100.00	3,000	141000.00 – 159000.00
600	38100.00 – 42600.00	5,000	228000.00 – 259000.00

LARGE SAWMILL OR PLANNING MILL – STEEL

Average costs for average heights, installed.

Base Area	Cost	Base Area	Cost
500 sq. ft.	25300.00	2,500 sq. ft.	100000.00
1,000	46200.00	3,000	116000.00
1,500	66000.00	4,000	150000.00
2,000	83000.00	5,000	180000.00

Add for forced draft systems:

6” - 7100.00	10” - 9850.00	15” - 13500.00
8” - 8650.00	12” - 11300.00	

PIPELINE COSTS

Moderate pressure, long run (over 5 miles in length), cross country, welded steel, underground oil and gas transmission lines, not including compressors, pumping stations, bridges, etc. Costs are smoothed averages of contract costs excluding extremes. The normal range is from 75% to 150% of the listed cost ranges, depending on length and type of pipe and pipe protection, terrain and geology, climate, location, etc.: e.g., the shorter the run, the more difficult, complex or urbanized the site, the higher the costs. Right-of-way costs are not included.

SIZE (DIAM.)	COST RANGE (PER MILE)		
	Low Cost	Avg.	Good
6”	377000.00	691000.00	1262000.00
8”	429000.00	733000.00	1303000.00
10”	454000.00	809000.00	1421000.00
12”	471000.00	849000.00	1521000.00
16”	681000.00	1202000.00	2160000.00
20”	1001000.00	1579000.00	2472000.00
24”	1202000.00	1950000.00	3101000.00
30”	1579000.00	2380000.00	3582000.00
36”	2203000.00	3052000.00	4203000.00
42”	2851000.00	3832000.00	5162000.00

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TRAILER AND MANUFACTURED HOUSING PARKS

SECTION UIP 10

INTRODUCTION

The park costs in this section are divided into five quality classifications and give a range from the cheap transient park to the highly developed manufactured housing park designed for permanent living. Many parks will be mixed in quality and may have good quality buildings, recreational facilities and low cost utilities or roads, or they may have few extra facilities and large patios but average roads. For these hybrid parks, the costs of the various items should be chosen from the quality of park where they would normally be found and built up to fit the subject park specifically. For example, a low cost park in a cold climate may have concrete paved homestands completely under the trailers. One could either price the patio and walk costs from the excellent category or divide the low-quality lump sum cost by the typical square footage and thus develop a square foot cost to apply to the specific average area of flatwork for the park in question.

The costs are for organized commercial parks and do not include the poorest resort types, which merely provide a parking space and some common facilities such as restrooms and a water source. The costs listed are medians for each classification, excluding extremes.

The costs are broken into major cost items on a cost-per-trailerspace basis, and miscellaneous costs such as the cost of normal financing and contractors' profit and overhead are prorated to each item. Architects' and designers' fees are included in the engineering costs for all items except buildings, whose costs will include all fees applicable to the structural improvements. Developers' overhead and profit, advertising and other promotional expenses are not included. Local jurisdictional fees or assessments are not included and must be obtained locally.

Off-site costs are not included. These may be costs of bringing utilities to the site, storm drains, access roads, bridges, traffic control, environmental impact studies, etc.

For hillside parks, the cost of grading and terracing the hillside sites must be added. Also, other costs such as paving, sewers and water will be higher for hillside installation. All excluded costs must be examined, as, in some cases, special local assessments and fees, off-site costs, developers' overhead and profit, etc., have equalled as much as an additional 40% of total project costs.

COST MODIFIERS

Costs for each park quality have been adjusted to a base number of units and gross area per unit which are roughly normal for that quality. Under the base costs for each quality, multipliers are given to adjust costs for deviations from the base. To determine the gross area per unit, divide the entire improved area of the park by the number of units.

DESCRIPTION OF COST ITEMS

ENGINEERING includes plans, engineering, permits and design and specifications of the park, exclusive of buildings.

GRADING includes leveling the site for drainage and roughing out

roads but does not include excavation and terracing for hillside sites. It will normally include some leveling of the trailer sites in the Average and Good qualities.

STREET PAVING includes base preparation and paving.

PATIOS AND WALKS include all flat work except street paving.

SEWER includes all on-site work and laterals but does not include sewage disposal systems, off-site connections to a trunk line, or connection charges, except for septic tanks and cesspools, which are included in the Cheap quality. Storm sewers are not included.

WATER includes all on-site mains, site services (laterals) and sprinkler systems but does not include wells, storage tanks, pumps, offsite connections to other sources or connection fees.

GAS includes all on-site piping and site connections (laterals) as well as connections to buildings, but does not include gas plumbing in buildings or off-site mains.

ELECTRICAL includes all on-site conduit, electrical and telephone wiring, trailer site outlets, street lighting commensurate with the quality, and area lighting. It does not include building lighting or offsite connections.

BUILDINGS include structures commensurate with the quality and size of the parks. It is often better to compute these from other sections of the manual, since the age and lifestyle of the occupants will dictate the type of amenities offered. For example, an upscale resort retirement project may have a large, high cost activities clubhouse, while first-time family park design may stress outdoor recreation with a smaller, very limited, low cost clubhouse facility. These costs also include all building design costs, and plumbing and electrical items for the buildings. Built-in appliances and all furnishings must be added separately.

MISCELLANEOUS includes an average amount of entrance ornamentation, signs and common landscaping commensurate with the park quality. Outdoor recreational facilities, swimpools, tennis courts, etc. and ornamental lakes and ponds should always be computed as extras. Recreational equipment, game tables, kitchen equipment, etc. are not included. Off site signs are not included.

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TRAILER AND RECREATIONAL VEHICLE PARKS

SECTION UIP 10

CHEAP

Typical sites developed for transient use in outlying rural or resort areas where there are either no building codes or minimal code enforcement. They will be closely spaced, have few facilities beyond the minimum subsistence level and be designed for smaller trailers and recreational vehicles. The base area per trailer space is 1,600 square feet and the base number of spaces is 50.

ENGINEERING – Minimum plans, engineering and permits.	$ 474.00
GRADING – Minimum leveling, graded for drainage, cleared.	405.00
STREET PAVING – Minimum asphalt, natural base, 15' to 20' wide, paved parking area	950.00
PATIOS AND WALKS – Average 135 square feet of small asphalt patio or hard-stand, few walks near buildings.	535.00
SEWER – 3" to 4" clay, few traps or vents. Cesspool and septic tank are included.	735.00
WATER – 2" mains, service to common hydrants and buildings, no trailer hookups.	585.00
GAS – None, except bottled gas (not included).	-----
ELECTRICAL – Low amperage circuits, overhead wiring, simple outlets at trailer sites. No telephones or speaker systems.	795.00
BUILDINGS – Restrooms and showers, laundry, office, lowest-cost frame or concrete block, cheap fixtures and partitions . . .	2140.00
MISCELLANEOUS – Signboard, minimum landscape and entrance	318.00
CHEAP PARK – Cost per space	6950.00

LOW COST

Typical sites developed for transient or semipermanent occupancy in seasonal resort areas or near industrial or military areas. Usually designed to hold car drawn trailers up to 40 to 45 feet long. The base area per site is 2,400 square feet and the base number of spaces is 80.

ENGINEERING – Limited plans and specifications, survey of site and permits	$ 920.00
GRADING – Graded for drainage, roads roughed in, site cleared and minimum site leveling	750.00
STREET PAVING – 18' to 22' roadway, 2" asphalt on natural base, no curbs or edging, common parking area	1490.00
PATIOS AND WALKS – Average 205 square feet of low cost concrete or asphalt for hardstand, patio or parking, some walks near buildings	960.00
SEWER – 4" lines, 6" mains, minimum code, simple layout	1080.00
WATER – 3" mains, 3/4" service, hydrant every two spaces	960.00
GAS – None to trailer spaces. Low pressure gas to utility buildings and office	452.00
ELECTRICAL – Overhead wiring, 30 to 80 amperes per space. Some street lights. Speaker system. Telephone booth (not included)	1430.00
BUILDINGS – Utility, showers and restrooms, laundry and office	2850.00
MISCELLANEOUS – Sign, low cost landscape, some masonry or concrete work around entrance. Swimming pool costs should be added from Section UIP 13.	765.00
LOW COST PARK – Cost per space	11600.00

MODIFIERS

NUMBER OF SPACES	10	20	30	40	50	60	80	100	120
MULTIPLIER	1.12	1.08	1.05	1.02	1.00	.98	.95	.93	.91

GROSS AREA PER SPACE	800	1,000	1,200	1,400	1,600	1,800	2,000	2,400	2,800
MULTIPLIER	.86	.89	.93	.96	1.00	1.03	1.06	1.12	1.28

MODIFIERS

NUMBER OF SPACES	30	40	50	50	60	80	100	140	160
MULTIPLIER	1.10	1.07	1.05	1.03	1.00	.97	.95	.93	.91

GROSS AREA PER SPACE	1,200	1,600	2,000	2,200	2,400	2,600	2,800	3,200	3,600
MULTIPLIER	.83	.89	.95	.97	1.00	1.02	1.05	1.09	1.12

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MANUFACTURED HOUSING PARKS

SECTION UIP 10

AVERAGE

These are average costs of parks built more or less for permanent occupancy and represent the low end midpoint for permanent parks. They will have spaces to accommodate the manufactured house up to 55 or 60 feet long, as well as large transient trailers. They will have utility buildings, offices, recreation buildings and other recreational facilities, which may be computed from other sections of the service. The base area per site is 3,200 square feet, and the base number of spaces is 100.

ENGINEERING – Well detailed plans, site survey, simple specifications, permits and bonds	$ 1300.00
GRADING – Graded for drainage, roads roughed in, some trailer site leveling	1120.00
STREET PAVING – 22' to 26' wide, 2" asphalt on good base, some edging or curb, some common parking	1930.00
PATIOS AND WALKS – Average 310 square feet of concrete for patios, hard-stands and walks near buildings	1420.00
SEWER – 4" service, 6" mains, adequate vents, good code installation	1580.00
WATER – 3" to 4" mains, valve connections and hydrants at sites	1280.00
GAS – Low-pressure gas to all sites and buildings	685.00
ELECTRICAL – Underground conduit, 60 to 100 amperes per site. Tele-phone extensions in buildings and most sites. Speaker system. Lighted recreation areas and street	2060.00
BUILDINGS – Utility, laundry, recreation, public restrooms	3650.00
MISCELLANEOUS – Adequate landscaping and sprinklers, some masonry ornamentation, average sign and entrance. Outdoor recreational facilities are not included and should be added from Sections UIP 13 and UIP 14.	1240.00
AVERAGE PARK – Cost per space	16300.00

GOOD

The typical good park is a manufactured housing park catering to the larger manufactured homes and represents the median for permanent parks. It will accommodate large manufactured homes with private patios and gardens, and offers complete recreational facilities. The base area per site is 4,400 square feet, and the base number of spaces is 175.

ENGINEERING – Complete detailed plans and specifications, permits, bonds and survey	$ 1890.00
GRADING – Graded for drainage, view, and appearance, roads roughed in	1720.00
STREET PAVING – Good 3" asphalt roadways on prepared base, 26' to 32' wide, edged or curbs, parking areas for visitors and extra cars	2725.00
PATIOS AND WALKS – Home stands, patios, and car stands. Average 465 square feet of concrete per space, including walks around buildings and recreation areas	2140.00
SEWER – 4" service, 6" mains, 8" trunk, good code installation, well vented and trapped	2000.00
WATER – 4" to 6" mains, good valve connections and hydrants at sites	1800.00
GAS – Low-pressure gas to all sites and buildings	1100.00
ELECTRICAL – Underground conduit, 80 to 150 amperes per space. Good street lighting and lighted recreation areas. Costs include telephone connection boxes at sites and cable TV systems	2975.00
BUILDINGS – Office, recreation, laundry	4650.00
MISCELLANEOUS – Above-average landscaping and sprinklers, signs, masonry ornamentation and walls. Outdoor recreational facilities are not included and should be added from Sections UIP 13 and 14.	2210.00
GOOD PARK – Cost per space	23200.00

MODIFIERS

NUMBER OF SPACES	40	60	80	100	125	150	175	200	250
MULTIPLIER	1.10	1.06	1.03	1.00	.97	.95	.93	.91	.89

GROSS AREA PER SPACE	2,000	2,400	2,800	3,200	3,600	4,000	4,400	4,800	5,200
MULTIPLIER	.89	.93	.97	1.00	1.03	1.05	1.08	1.10	1.12

MODIFIERS

NUMBER OF SPACES	50	75	100	125	150	175	200	250	300
MULTIPLIER	1.17	1.12	1.08	1.05	1.02	1.00	.98	.96	.95

GROSS AREA PER SPACE	2,800	3,200	3,600	4,000	4,400	4,800	5,200	5,600	6,000
MULTIPLIER	.91	.93	.96	.98	1.00	1.01	1.03	1.05	1.06

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MANUFACTURED HOUSING PARKS

SECTION UIP 10

EXCELLENT

The excellent manufactured housing park provides deluxe accommodations for the largest site erected manufactured home units and represents the high end midpoint for permanent parks. It will have complete and varied recreational facilities of top quality. The base area per site is 5,600 square feet, and the base number of spaces is 200.

ENGINEERING – Complete detailed plans and specifications, permits, bonds and survey	$ 2500.00
GRADING – Graded for drainage, view, and appearance, roads roughed in.	2390.00
STREET PAVING – Good 3" asphalt roadways on prepared base, 32' to 40' wide, curbs, finished parking areas for visitors and extra cars.	3600.00
PATIOS AND WALKS – Home stands, patios, and car stands. Average 700 square feet of concrete per space, including walks around buildings and recreation areas	3150.00
SEWER – 4" service, 6" mains, 8" trunk, good code installation, well vented and trapped	2390.00
WATER – 4" to 6" mains, good valve connections and hydrants at sites	2360.00
GAS – Low-pressure gas to all home sites and buildings	1520.00
ELECTRICAL – Underground conduit, 100 to 200 amperes per space. Good street lighting, floodlighted recreation areas. Costs include telephone connection boxes at sites and cable TV systems	3975.00
BUILDINGS – Office, recreation, arts and crafts, laundromat	5250.00
MISCELLANEOUS – Generous amounts of landscaping and sprinklers, large signs, masonry ornamentation and walls. Outdoor recreational facilities and ornamental lakes and ponds should be added from Sections UIP 13 and 14.	3175.00
EXCELLENT PARK – Cost per space	30200.00

MODIFIERS

NUMBER OF SPACES	50	100	150	175	200	225	250	300	350
MULTIPLIER	1.18	1.10	1.04	1.02	1.00	.99	.98	.97	.97

GROSS AREA PER SPACE	4,000	4,400	4,800	5,200	5,600	6,000	6,400	6,800	7,200
MULTIPLIER	.95	.97	.98	.99	1.00	1.01	1.01	1.02	1.03

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